

SAPIENZA
UNIVERSITÀ DI ROMA

Integer Linear Programming

Volume 1 — Models, bounds and formulations

What an integer linear programming problem is (MIP, ILP, BIP), its solutions and outcomes; primal and dual bounds and the gap; the LP relaxation; ideal formulations, valid inequalities and cover inequalities. Worked examples and exercises with complete solutions.

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Part I

Models and bounds

Chapter 1

Integer linear programming

Prerequisites

- From the module *Linear Programming*: Chapter 2 «Halfspaces and polyhedra» and Chapter 4 «LP problems» (the canonical maximization form, the feasible region, optimal solutions and the optimal value), and Chapter 5 «Outcomes and equivalences» (infeasible, unbounded and optimal problems).
- Nothing else: this chapter is the starting point of the module. The modelling of real problems as MIP problems is the subject of the module *MIP modelling*.

Objectives

At the end of this chapter you will be able to:

- write an ILP problem in canonical maximization form, and distinguish ILP, BIP and MIP problems;
- draw the feasible region of a small ILP problem and find an optimal solution graphically;
- explain why the feasible region is not convex, and what the three outcomes of an ILP problem are;
- explain why complete enumeration fails, and how much it costs;
- compute a lower bound on the optimal value from a feasible solution, also with a simple heuristic algorithm.

1.1 Why integer linear programming?

- Many decisions are **discrete**: how many cars to produce, whether to open a warehouse, which job goes on which machine. The number of possible plans is finite, but it can be huge.
- An **integer linear programming** problem describes such a decision with linear constraints and a linear objective function, as in linear programming, but some variables must take integer values. A variable restricted to $\{0, 1\}$ is a yes/no decision (*in* or *out*), and it is the most powerful modelling tool of the module.
- The integrality requirement makes the problem much harder than its continuous version. The whole module answers one question: **how do we find an optimal solution, and how do we certify that it is optimal?**

A solution is certified optimal when a **lower bound** (a feasible solution) and an **upper bound** (a proof that nothing is better) have the same value. Every method of this module produces one of the two bounds, or both.

1.2 A first model: the car example

- A company produces two types of cars, **sports cars** (profit 3 thousand euros per car) and **family cars** (profit 2 thousand euros per car). Every week it has 45 hours of labour and 49 units of special materials. The consumption of each car is the following:

	Sports car	Family car
Labour (hours)	10	4
Special materials (units)	4	14

- **Decision variables**: x_1 is the number of sports cars and x_2 the number of family cars produced in a week. Cars cannot be split, so both variables are non-negative integers.
- **Objective function**: the weekly profit $3\,x_1 + 2\,x_2$ (in thousands of euros), to be maximized.
- **Constraints**: the labour hours used, $10\,x_1 + 4\,x_2$, cannot exceed 45, and the units of materials used, $4\,x_1 + 14\,x_2$, cannot exceed 49. The ILP problem reads:

$$\begin{aligned} \max \quad & 3\,x_1 + 2\,x_2 & & \\ \text{subject to} \quad & 10\,x_1 + 4\,x_2 & \leq & \ 45 \\ & 4\,x_1 + 14\,x_2 & \leq & \ 49 \\ & x_1, \ x_2 & \in & \ \mathbb{Z}_{\geq 0} \end{aligned}$$

- **A challenge.** Before reading on, find a good production plan. Then try to answer the harder question: how can you prove that no better plan exists? The car example is the guiding example of the whole module, and each chapter will give a piece of the answer.

1.3 Definitions

1.3.1 ILP problems in canonical maximization form

- Given n values $c_j \in \mathbb{Z}$, $m \cdot n$ values $a_{ij} \in \mathbb{Z}$ and m values $b_i \in \mathbb{Z}$, an **integer linear programming** (ILP) problem in **canonical maximization form** with n **non-negative integer variables** and m **less-than-or-equal-to inequality constraints** reads as follows:

$$\max \quad \sum_{j=1}^{n} c_j \, x_j \tag{1.1a}$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} \, x_j \leq b_i, \qquad \forall i \in \{1, 2, \ldots, m\}, \tag{1.1b}$$

$$x_j \in \mathbb{Z}_{\geq 0}, \qquad \forall j \in \{1, 2, \ldots, n\}. \tag{1.1c}$$

- The car example is an ILP problem with $n = 2$ and $m = 2$.
- If all the variables are restricted to $\{0, 1\}$ we have a **binary integer linear programming** (BIP) problem. If only some variables are integer and the others are continuous we have a **mixed integer linear programming** (MIP) problem. The ILP problem is the *pure* integer case; this chapter and the next ones treat ILP problems, and the MIP case is mentioned only where something changes.
- Constraints $\geq$ and $=$, and minimization problems, are brought to the canonical form as in linear programming: multiply by -1 the constraints $\geq$, replace an equality with two inequalities, and maximize $-f$ instead of minimizing f.

1.3.2 Feasible region, objective function and optimal solution

- The **feasible region** of the ILP problem (1.1) is the following **non-convex** set:

$$F = \left\{ \boldsymbol{x} \in \mathbb{Z}^n : \underbrace{\sum_{j=1}^{n} a_{ij} \, x_j \leq b_i, \ \forall i}_{m \text{ inequality constraints}} , \quad \underbrace{x_j \geq 0, \ \forall j}_{n \text{ non-negative variables}} \right\}$$

  It is the set of the integer points (the **lattice** $\mathbb{Z}^n$) that lie in the polyhedron defined by the $m + n$ half-spaces of the constraints. Since F is a set of isolated points, it is not convex.

- The **objective function** (1.1a) is the scalar linear function f defined as follows:

$$f : F \to \mathbb{R}, \quad \boldsymbol{x} \mapsto \sum_{j=1}^{n} c_j \, x_j$$

  whose domain is the set F.

Solving an ILP problem (1.1) requires to determine the **global maximum** $z(\text{ILP})$ of the objective function over the feasible region (if it exists). The global maximum $z(\text{ILP})$ is called the **optimal value** of the ILP problem (1.1).

Solving an ILP problem (1.1) also requires to determine a **point of global maximum**:

$$\tilde{\boldsymbol{x}}(\text{ILP}) = \arg\max \left\{ \; f(\boldsymbol{x}) : \; \boldsymbol{x} \in F \; \right\}$$

which is called an **optimal solution** of the ILP problem (1.1). If an optimal solution $\tilde{\boldsymbol{x}}(\text{ILP}) = (\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$ exists, we have:

$$\sum_{j=1}^{n} c_j \, \tilde{x}_j = z(\text{ILP})$$

- We denote by $\bar{\boldsymbol{x}}$ a **feasible solution** of the ILP problem (1.1), that is, a point $\bar{\boldsymbol{x}} \in F$. An optimal solution is a feasible solution, but a feasible solution is in general not optimal.

1.3.3 The three outcomes of an ILP problem

- As in linear programming, an ILP problem has exactly one of three **outcomes**: it is **infeasible** ($F = \emptyset$), **unbounded** (the objective function is not bounded from above on F) or it has an **optimal solution**.

- The integrality requirement changes the outcomes in a subtle way. The problem $\max\{x_1 : 2\,x_1 \leq 1, \; -2\,x_1 \leq -1, \; x_1 \in \mathbb{Z}_{\geq 0}\}$ is infeasible, because it requires $x_1 = \frac{1}{2}$, although its LP relaxation (without integrality) is feasible.

- If the polyhedron of the constraints is bounded, then F is a **finite set**, and an optimal solution exists as soon as F is not empty. If the polyhedron is unbounded, F can be infinite: for example, the problem $\max\{x_1 + x_2 : x_1 - x_2 \leq 1, \; x_1, x_2 \in \mathbb{Z}_{\geq 0}\}$ is unbounded.

1.4 Feasible and optimal solutions of the car example

- The feasible solutions of the car example are the integer points inside the polyhedron defined by the two constraints and by $x_1, x_2 \geq 0$. There are 16 of them.

- To find an optimal solution graphically, we move the objective line $3\,x_1 + 2\,x_2 = z$ in the direction of increasing z until it touches the last feasible point. This happens at the point $(4, 1)$.

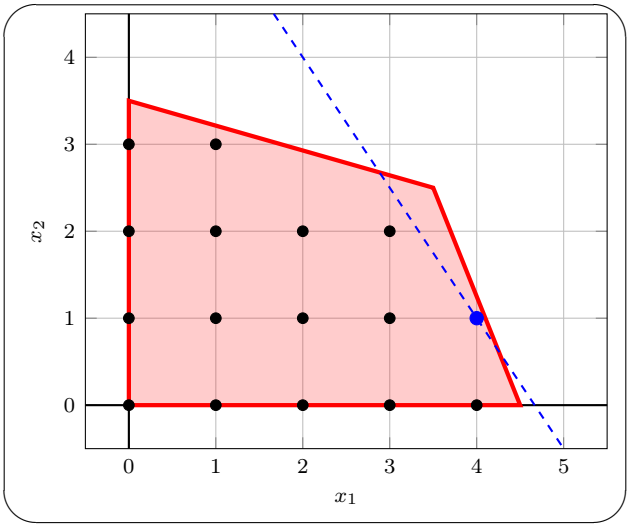


Figure 1.1: The car example: the feasible region of the ILP problem is the set of its 16 integer points (black) inside the polyhedron (red). The optimal solution $(4, 1)$ is blue, with the objective line $3\,x_1 + 2\,x_2 = 14$ (dashed).

The car example has the **unique** optimal solution $\tilde{\boldsymbol{x}}(\text{ILP}) = (4, 1)$ with optimal value $z(\text{ILP}) = 14$: the company produces 4 sports cars and 1 family car per week, for a profit of 14 thousand euros.

- The point $\bar{\boldsymbol{x}} = (2, 2)$ is feasible, because $10 \cdot 2 + 4 \cdot 2 = 28 \leq 45$, $4 \cdot 2 + 14 \cdot 2 = 36 \leq 49$ and both values are integer. The point $(4, 2)$ is not feasible (it uses 48 hours of labour), and neither is $(\frac{7}{2}, \frac{5}{2})$, which is not integer.

1.5 Complete enumeration

- When the feasible region is a finite set we can find an optimal solution by **complete enumeration**: we compute the objective function for every feasible solution and keep the best. For the car example, there are 16 points:

x_1	x_2	z	x_1	x_2	z
0	0	0	3	1	11
1	0	3	**4**	**1**	**14**
2	0	6	0	2	4
3	0	9	1	2	7
4	0	12	2	2	10
0	1	2	3	2	13
1	1	5	0	3	6
2	1	8	1	3	9

- The best value is 14, in $(4, 1)$ (in bold). Here the method works, because the model is tiny.

- Consider now the simplest case, n **binary variables** and no other constraint. Every variable takes 2 values, so there are 2^n solutions: the vertices of an n-dimensional hypercube.

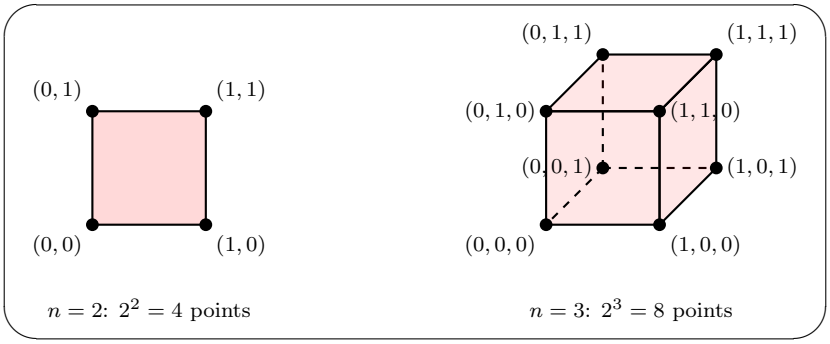


Figure 1.2: The solutions of n binary variables without other constraints: the 2^n vertices of an n-dimensional hypercube, for $n = 2$ (a square) and $n = 3$ (a cube).

- The number 2^n explodes. With $n = 80$ variables there are about $1.2 \cdot 10^{24}$ solutions, comparable to the estimated number of stars in the universe, and with $n = 266$ variables about $1.2 \cdot 10^{80}$, comparable to the estimated number of atoms in the observable universe.

- Suppose that a computer examines 10^9 solutions per second, and then a computer 1000 times faster, 10^{12} solutions per second. The time needed to enumerate 2^n solutions is:

n	2^n	at 10^9 solutions/s	at 10^{12} solutions/s
20	$\approx 1.05 \cdot 10^6$	≈ 1.05 ms	$\approx 1.05\,\mu$s
30	$\approx 1.07 \cdot 10^9$	≈ 1.07 s	≈ 1.07 ms
40	$\approx 1.10 \cdot 10^{12}$	≈ 18.3 min	≈ 1.10 s
50	$\approx 1.13 \cdot 10^{15}$	≈ 13.0 days	≈ 18.8 min
60	$\approx 1.15 \cdot 10^{18}$	≈ 36.5 years	≈ 13.3 days
70	$\approx 1.18 \cdot 10^{21}$	≈ 374 centuries	≈ 37.4 years
80	$\approx 1.21 \cdot 10^{24}$	$\approx 3.8 \cdot 10^5$ centuries	≈ 383 centuries

Complete enumeration is impractical already for models of moderate size. A computer 1000 times faster gains only about 10 variables, because $2^{10} = 1024 \approx 1000$.

- The way out is to **avoid enumerating**. If we know a lower bound and an upper bound on the optimal value, we can exclude entire regions of feasible solutions without visiting them. The two bounds come from two different sources, and we start with the lower bound.

1.6 Lower bounds from feasible solutions

- Since the problem is a maximization, every feasible solution $\bar{\boldsymbol{x}}$ gives a **lower bound** on the optimal value:
$$\text{LB} = \sum_{j=1}^{n} c_j \, \bar{x}_j \leq z(\text{ILP}).$$
- For the feasible solution $(2, 2)$ of the car example we get $\text{LB} = 3 \cdot 2 + 2 \cdot 2 = 10$, so $z(\text{ILP}) \geq 10$.

A good feasible solution is a good lower bound. A **heuristic algorithm** is a method that finds feasible solutions quickly, with no guarantee of optimality.

1.6.1 Two heuristic algorithms for the car example

- The **one-type heuristic** produces the maximum number of cars of a single type, with none of the other type, and keeps the better of the two solutions.
  1. Only sports cars ($x_2 = 0$): $\bar{x}_1 = \min\{\lfloor \frac{45}{10} \rfloor, \lfloor \frac{49}{4} \rfloor\} = \min\{4, 12\} = 4$, so $\bar{\boldsymbol{x}} = (4, 0)$ and $\text{LB} = 12$.
  2. Only family cars ($x_1 = 0$): $\bar{x}_2 = \min\{\lfloor \frac{45}{4} \rfloor, \lfloor \frac{49}{14} \rfloor\} = \min\{11, 3\} = 3$, so $\bar{\boldsymbol{x}} = (0, 3)$ and $\text{LB} = 6$.
- The **two-step heuristic** produces first the maximum number of cars of one type, and then the maximum number of cars of the other type with the **residual resources**. It keeps the better of the two solutions.
  1. Sports cars first: $\bar{x}_1 = 4$ uses 40 hours and 16 units, leaving 5 and 33. Then $\bar{x}_2 = \min\{\lfloor \frac{5}{4} \rfloor, \lfloor \frac{33}{14} \rfloor\} = \min\{1, 2\} = 1$, so $\bar{\boldsymbol{x}} = (4, 1)$ and $\text{LB} = 14$.
  2. Family cars first: $\bar{x}_2 = 3$ uses 12 hours and 42 units, leaving 33 and 7. Then $\bar{x}_1 = \min\{\lfloor \frac{33}{10} \rfloor, \lfloor \frac{7}{4} \rfloor\} = \min\{3, 1\} = 1$, so $\bar{\boldsymbol{x}} = (1, 3)$ and $\text{LB} = 9$.

Heuristic	Order	$\bar{x}_1$	$\bar{x}_2$	LB
one-type	sports cars	4	0	12
one-type	family cars	0	3	6
two-step	sports cars first	4	1	14
two-step	family cars first	1	3	9

- The two-step heuristic gives a better lower bound, but it needs more computation. This trade-off between the quality of the solution and the computational effort is typical of heuristic algorithms.
- Here the best lower bound is $\text{LB} = 14$, in $(4, 1)$, which is in fact optimal. But we cannot know it yet: nothing so far tells us that no feasible solution has value 15 or more. We need an **upper bound**, and the next chapter provides it.

1.7 Typical mistakes

Warning

- **Believing that the feasible region is convex.** It is the set of the integer points of a polyhedron, not the polyhedron. The midpoint of two feasible solutions can be infeasible.
- **Believing that the feasible region is always finite.** It is finite only if the polyhedron is bounded; otherwise the ILP problem can be unbounded.
- **Believing that a feasible LP problem gives a feasible ILP problem.** The LP problem can have solutions while no integer point satisfies the constraints.
- **Taking a heuristic solution as optimal.** A feasible solution gives only a lower bound. Without an upper bound with the same value there is no certificate of optimality.
- **Using the lower bound as an upper bound.** For a maximization problem the feasible solutions bound the optimal value from below, and the relaxations (next chapters) from above.

1.8 In summary

In summary

- An ILP problem maximizes a linear function over the integer points of a polyhedron; its feasible region is not convex, and it is finite when the polyhedron is bounded.
- The outcomes are infeasible, unbounded or optimal, and a feasible LP problem can have an infeasible ILP version.
- Complete enumeration costs 2^n evaluations for n binary variables: it is not an option beyond very small models.
- Every feasible solution is a lower bound on $z(\text{ILP})$, and heuristic algorithms find good feasible solutions quickly.
- Optimality is certified only when a lower bound and an upper bound coincide.

Dashboard of the car example. The bounds on $z(\text{ILP})$ found so far:

Chapter	Lower bound	Upper bound	From
Integer linear programming	12, 14	–	heuristic solutions $(4, 0)$ and $(4, 1)$

Chapter 2

LP relaxation and dual bounds

Prerequisites

- From this module, the chapter «Integer linear programming»: ILP problems, feasible and optimal solutions, lower bounds, and the car example.
- From the module *Linear Programming*, Chapter 9 «Duality theory»: the dual problem, weak duality and strong duality.
- From the same module, Chapter 6 «Simplex algorithm»: an LP problem that has an optimal solution has an optimal vertex.

Objectives

At the end of this chapter you will be able to:

- build the LP relaxation of an ILP problem and use its optimal value as an upper bound, rounded down when it is valid;
- compute the optimality gap from a lower bound and an upper bound;
- explain why rounding an optimal LP solution gives a feasible solution at best;
- derive upper bounds on the variables from the constraints, and say when they help;
- obtain an upper bound from non-negative multipliers of the constraints, and recognize the dual problem as the problem of the best multipliers.

Where we were

In the previous chapter we found the feasible solution $(4, 1)$ of the car example, with LB $= 14$, but we could not tell whether a plan with a larger profit exists. We now look for the other side of the sandwich, an **upper bound**.

2.1 Why a relaxation?

- To prove that no production plan earns more than 14 thousand euros we cannot look at the solutions one by one. We need a statement that holds for all the feasible solutions at once.
- The idea is to **relax** the problem: we replace it with an easier problem whose feasible region contains the original one. A maximization over a larger region cannot give a smaller value, so the optimal value of the relaxed problem is an upper bound.
- The easier problem we choose is the one we know how to solve: a linear programming problem.

2.2 The LP relaxation

- The **linear programming (LP) relaxation** of an ILP problem consists in replacing the integrality constraints with the non-negativity constraints $x_j \geq 0$, for all $j \in \{1, 2, \dots, n\}$. In other words, we transform the integer variables into continuous variables, and we get the following problem:

$$\max \sum_{j=1}^{n} c_j x_j \tag{2.1a}$$

$$\text{subject to} \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \qquad \forall i \in \{1, 2, \dots, m\}, \tag{2.1b}$$

$$x_j \geq 0, \qquad \forall j \in \{1, 2, \dots, n\}. \tag{2.1c}$$

- We denote by $z(\text{LP})$ the optimal value of the LP relaxation, and by $\tilde{\boldsymbol{x}}(\text{LP})$ an optimal solution. We always write the argument, because in the same example $\tilde{\boldsymbol{x}}(\text{ILP})$ and $\tilde{\boldsymbol{x}}(\text{LP})$ are different points.

Proposition 2.1: upper bound from the LP relaxation

Every feasible solution of the ILP problem is feasible for its LP relaxation, hence

$$z(\text{ILP}) \leq z(\text{LP}). \tag{2.2}$$

Moreover, if an optimal solution of the LP relaxation is integer, then it is an optimal solution of the ILP problem.

Proof. The feasible region F of the ILP problem is contained in the polyhedron of the LP relaxation, and the objective function is the same. The maximum over a larger set is not smaller than the maximum over a subset. If $\tilde{\boldsymbol{x}}(\text{LP})$ is integer, it belongs to F and has value $z(\text{LP}) \geq z(\text{ILP})$: it is optimal for the ILP problem. □

Definition 2.2: of integer part of a number

Given $y \in \mathbb{R}$, the **integer part** of y is $\lfloor y \rfloor = \max\{q \in \mathbb{Z} : q \leq y\}$.

- The value of every feasible solution of an ILP problem with integer costs c_j is an integer. Since it is at most $z(\text{LP})$, it is at most the largest integer not exceeding $z(\text{LP})$.

Observation 2.3: rounding the upper bound

If all the variables are integer and all the costs c_j are integer, then

$$z(\text{ILP}) \leq \lfloor z(\text{LP}) \rfloor. \tag{2.3}$$

In all the other cases the bound $\lfloor z(\text{LP}) \rfloor$ is not valid.

- In this module all the ILP problems have integer data, so the bound (2.3) always holds. The case of a MIP problem is different: if a variable is continuous, the value of a feasible solution is not necessarily an integer, and the bound cannot be rounded.

2.2.1 The LP relaxation of the car example

- The LP relaxation of the car example has the optimal solution

$$\tilde{x}_1 = \frac{7}{2}, \quad \tilde{x}_2 = \frac{5}{2}, \qquad z(\text{LP}) = \frac{31}{2},$$

  and it is fractional: it is not a production plan.

- Since the costs are integer, we have the upper bound $\lfloor z(\text{LP}) \rfloor = 15$. Together with the lower bound of the previous chapter:

$$14 \ \leq \ z(\text{ILP}) \ \leq \ 15.$$

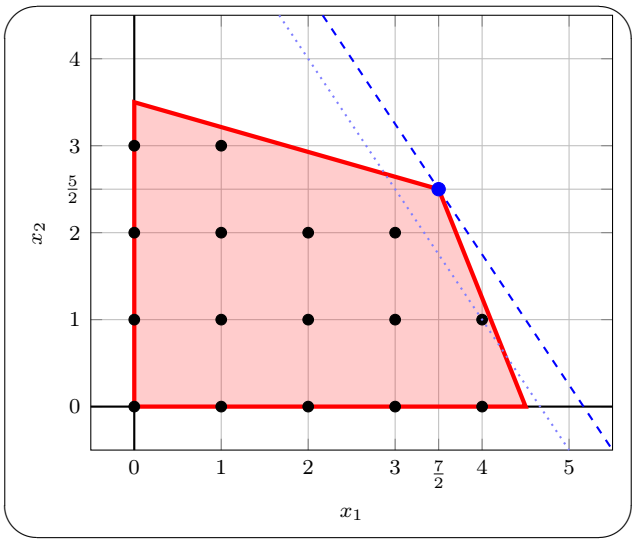


Figure 2.1: The LP relaxation of the car example: the feasible region (red), its 16 integer points (black) and the optimal solution $(\frac{7}{2}, \frac{5}{2})$ of the LP relaxation (blue), with the objective line $3\,x_1 + 2\,x_2 = \frac{31}{2}$ through it (dashed). The dotted line is the objective line of the optimal ILP solution.

- We know that the optimal value is 14 or 15. The **optimality gap** measures how far the two bounds are:

$$\text{optimality gap} = \frac{\text{UB} - \text{LB}}{|\text{LB}|}, \qquad \text{for LB} \neq 0. \tag{2.4}$$

  Here UB = 15 and LB = 14, so the gap is $\frac{1}{14} \approx 7.1\%$. The solution $(4, 1)$ is certified optimal only when the gap is zero.

2.3 Rounding the solution of the LP relaxation

Feasible ILP solutions can typically be obtained by rounding an optimal solution of the LP relaxation. But no rounding technique guarantees to find an optimal ILP solution.

Example 2.1: rounding works, but the bound is not tight

Consider the following ILP problem; its LP relaxation has $x_1, x_2 \geq 0$ instead of the integrality constraints:

$$\begin{array}{rrcrcr} \max & 2\,x_1 & + & 4\,x_2 & & \\ \text{subject to} & -\,x_1 & + & x_2 & \leq & 2 \\ & 8\,x_1 & + & 2\,x_2 & \leq & 19 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

- The optimal solution of the LP relaxation is $\tilde{\boldsymbol{x}}(\text{LP}) = (\frac{3}{2}, \frac{7}{2})$, with $z(\text{LP}) = 17$ and $\lfloor z(\text{LP}) \rfloor = 17$, so $z(\text{ILP}) \leq 17$.
- Rounding down both components we get the feasible solution $\bar{\boldsymbol{x}} = (\lfloor \frac{3}{2} \rfloor, \lfloor \frac{7}{2} \rfloor) = (1, 3)$, with $\text{LB} = 2 \cdot 1 + 4 \cdot 3 = 14$.
- The optimal ILP solution is $\tilde{\boldsymbol{x}}(\text{ILP}) = (1, 3)$, with $z(\text{ILP}) = 14$: the rounded solution is optimal. But the bounds $14 \leq z(\text{ILP}) \leq 17$ do not prove it.

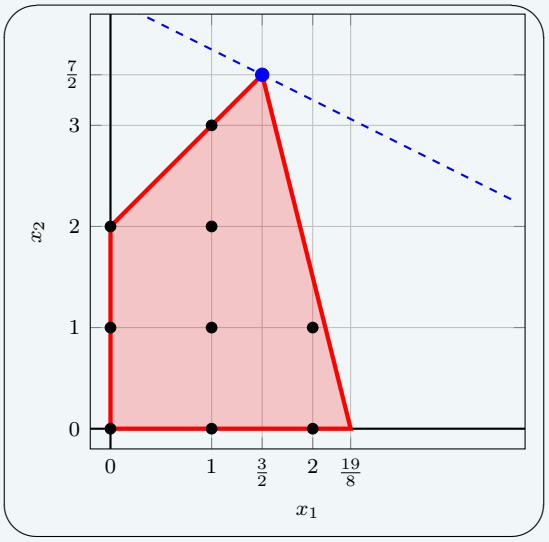


Figure 2.2: The second example: the feasible region of the LP relaxation (red), the integer points (black) and the optimal solution of the LP relaxation (blue), with the objective line $2\,x_1 + 4\,x_2 = 17$ (dashed).

Example 2.2: rounding fails

Consider the following ILP problem:

$$\begin{array}{rrcrcr} \max & 25\,x_1 & + & 16\,x_2 & & \\ \text{subject to} & 50\,x_1 & + & 31\,x_2 & \leq & 250 \\ & -3\,x_1 & + & 2\,x_2 & \leq & 4 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

- The optimal solution of the LP relaxation is $\tilde{\boldsymbol{x}}(\text{LP}) = (\frac{376}{193}, \frac{950}{193}) \approx (1.94, 4.92)$, with $z(\text{LP}) = \frac{24600}{193} \approx 127.46$. Hence $z(\text{ILP}) \leq \lfloor z(\text{LP}) \rfloor = 127$.
- Rounding to the nearest integer gives $(2, 5)$, which is infeasible: $50 \cdot 2 + 31 \cdot 5 = 255 > 250$. Rounding $\tilde{x}_1$ up and $\tilde{x}_2$ down gives the feasible solution $\bar{\boldsymbol{x}} = (2, 4)$, with $\text{LB} = 25 \cdot 2 + 16 \cdot 4 = 114$.
- The optimal ILP solution is $\tilde{\boldsymbol{x}}(\text{ILP}) = (5, 0)$, with $z(\text{ILP}) = 125$. It is far from $\tilde{\boldsymbol{x}}(\text{LP})$, and no rounding of $\tilde{\boldsymbol{x}}(\text{LP})$ finds it.

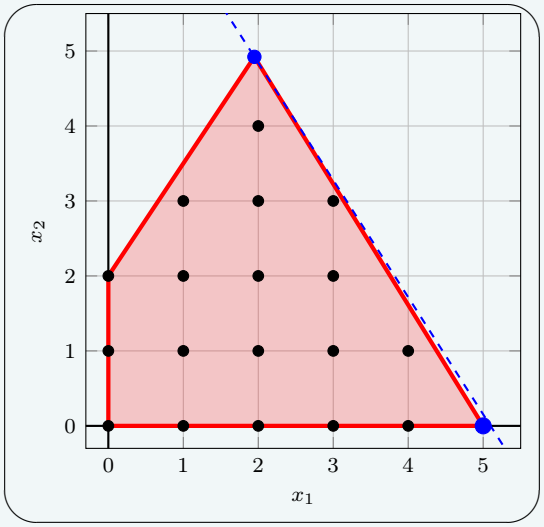


Figure 2.3: The third example: the integer points (black) are far from the optimal solution $(\frac{376}{193}, \frac{950}{193})$ of the LP relaxation (blue). The optimal ILP solution is $(5, 0)$ (blue circle). The dashed line is the objective line $25\,x_1 + 16\,x_2 = \frac{24600}{193}$.

Optimal ILP solutions can be quite far from optimal solutions of the LP relaxation. The value of the LP relaxation, instead, is often a good bound: here $125 \leq z(\text{LP}) \approx 127.46$.

2.4 Upper bounds on the variables

- Some constraints give a bound on a single variable. If a constraint has all the coefficients non-negative, then each term is non-negative, so no term can exceed the right-hand side.

Proposition 2.4: upper bounds on the variables

Let $\sum_{j=1}^{n} a_{ij}\, x_j \le b_i$ be a constraint with $a_{ij} \ge 0$ for all j. Then every feasible solution of the ILP problem satisfies

$$x_j \le \left\lfloor \frac{b_i}{a_{ij}} \right\rfloor, \qquad \forall j \text{ such that } a_{ij} > 0. \tag{2.5}$$

Proof. Since all the variables are non-negative and all the coefficients are non-negative, $a_{ij}\, x_j \le \sum_{k=1}^{n} a_{ik}\, x_k \le b_i$, so $x_j \le \frac{b_i}{a_{ij}}$. The variable is integer, so it does not exceed the integer part of $\frac{b_i}{a_{ij}}$. □

- If several constraints give a bound on x_j, we keep the smallest. In the car example both constraints have non-negative coefficients:

$$x_1 \le \min\left\{\left\lfloor \tfrac{45}{10} \right\rfloor, \left\lfloor \tfrac{49}{4} \right\rfloor\right\} = \min\{4, 12\} = 4,$$
$$x_2 \le \min\left\{\left\lfloor \tfrac{45}{4} \right\rfloor, \left\lfloor \tfrac{49}{14} \right\rfloor\right\} = \min\{11, 3\} = 3.$$

- The inequalities $x_1 \le 4$ and $x_2 \le 3$ are satisfied by all the integer points of the car example, and they can be added to the model. They cut off two regions of the polyhedron, and no integer point.

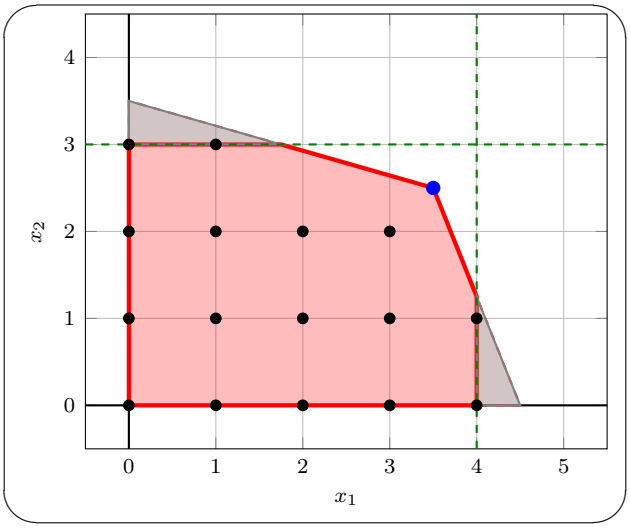


Figure 2.4: The variable upper bounds $x_1 \le 4$ and $x_2 \le 3$ (green) cut off the two gray regions of the LP relaxation (red) and no integer point. The optimal solution of the LP relaxation (blue) is not cut off.

- The optimal solution of the LP relaxation, $(\frac{7}{2}, \frac{5}{2})$, satisfies both inequalities. It is not cut off, so the optimal value of the LP relaxation does not change, and the upper bound is still 15.

- The bounds help when the optimal LP solution lies in a cut-off region. Consider the objective function $x_1 - x_2$ on the same constraints, whose optimal ILP solution is $(4, 0)$ with $z(\text{ILP}) = 4$. Before the bounds, the optimal solution of the LP relaxation is $(\frac{9}{2}, 0)$ with $z(\text{LP}) = \frac{9}{2}$; after $x_1 \leq 4$ it is $(4, 0)$ with $z(\text{LP}) = 4$, which is integer and therefore optimal.

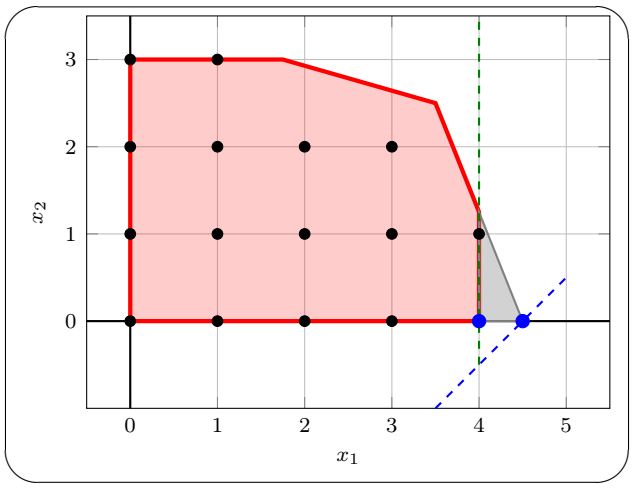


Figure 2.5: The objective $x_1 - x_2$: the variable upper bound $x_1 \leq 4$ (green) cuts off the gray triangle and the optimal solution $(\frac{9}{2}, 0)$ of the LP relaxation (blue circle). The new optimal solution $(4, 0)$ (blue) is integer. The dashed line is $x_1 - x_2 = \frac{9}{2}$.

- In this case the value of the LP relaxation improves from $\frac{9}{2}$ to 4. The rounded bound $\lfloor \frac{9}{2} \rfloor = 4$ was already 4: a better formulation does not always improve the rounded bound.

2.5 Dual bounds

- The LP relaxation gives a bound only after it has been solved. Can we obtain an upper bound without solving it? A feasible solution of the LP relaxation does not help: its value is neither a lower bound nor an upper bound on $z(\text{ILP})$.

(x_1, x_2)	feasible for the LP relaxation	integer	$3\,x_1 + 2\,x_2$
$(4, \frac{5}{4})$	yes	no	$\frac{29}{2} > 14$
$(2, \frac{5}{2})$	yes	no	$11 < 14$

- The right tool is to **combine the constraints**. Multiplying the first constraint of the car example by $\frac{3}{4}$ we get

$$\frac{15}{2}\,x_1 + 3\,x_2 \leq \frac{135}{4}.$$

Since $x_1, x_2 \geq 0$ and the coefficients $\frac{15}{2}$ and 3 are at least the costs 3 and 2, we have $3\,x_1 + 2\,x_2 \leq \frac{15}{2}\,x_1 + 3\,x_2 \leq \frac{135}{4}$: an upper bound of $\frac{135}{4}$.

- The same works with any **non-negative** multipliers π_1 and π_2 of the two constraints, provided that the coefficients of the sum are at least the costs:

$$(10\,\pi_1 + 4\,\pi_2)\,x_1 + (4\,\pi_1 + 14\,\pi_2)\,x_2 \le 45\,\pi_1 + 49\,\pi_2,$$
$$10\,\pi_1 + 4\,\pi_2 \ge 3, \qquad 4\,\pi_1 + 14\,\pi_2 \ge 2.$$

  The multipliers must be non-negative: a negative multiplier would reverse the inequality.

(π_1, π_2)	valid multipliers	upper bound $45\,\pi_1 + 49\,\pi_2$
$(\frac{3}{4}, 0)$	yes	$\frac{135}{4}$
$(0, \frac{3}{4})$	yes	$\frac{147}{4}$
$(\frac{1}{5}, \frac{1}{4})$	yes	$\frac{85}{4}$
$(\frac{1}{2}, 0)$	yes	$\frac{45}{2}$
$(\frac{1}{5}, 0)$	no: $10 \cdot \frac{1}{5} = 2 < 3$	–

- To get the **best** bound we look for the multipliers with the smallest value $45\,\pi_1 + 49\,\pi_2$: this is itself an LP problem.

$$\begin{aligned} \min \quad & 45\,\pi_1 + 49\,\pi_2 \\ \text{subject to} \quad & 10\,\pi_1 + 4\,\pi_2 \ge 3 \\ & 4\,\pi_1 + 14\,\pi_2 \ge 2 \\ & \pi_1, \pi_2 \ge 0 \end{aligned}$$

- This is the **dual problem** of the LP relaxation, called D(LP), and its optimal value is $z(\text{D(LP)})$. For a general ILP problem in canonical form, with one multiplier $\pi_i \ge 0$ for each constraint, it reads:

$$\min \quad \sum_{i=1}^{m} b_i\,\pi_i \tag{2.6a}$$

$$\text{subject to} \quad \sum_{i=1}^{m} a_{ij}\,\pi_i \ge c_j, \qquad \forall j \in \{1, 2, \dots, n\}, \tag{2.6b}$$

$$\pi_i \ge 0, \qquad \forall i \in \{1, 2, \dots, m\}. \tag{2.6c}$$

Proposition 2.5: dual bounds

For every feasible solution $\bar{\boldsymbol{\pi}}$ of the dual problem (2.6),

$$z(\text{ILP}) \;\le\; z(\text{LP}) \;\le\; \sum_{i=1}^{m} b_i\,\bar{\pi}_i.$$

By strong duality, $z(\mathrm{D(LP)}) = z(\mathrm{LP})$: the best multipliers give exactly the value of the LP relaxation.

Proof. Let $\boldsymbol{x}$ be feasible for the LP relaxation. Multiplying the constraint i by $\bar{\pi}_i \geq 0$ and summing, $\sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij}\,\bar{\pi}_i\right) x_j \leq \sum_{i=1}^{m} b_i\,\bar{\pi}_i$. By (2.6b) and $x_j \geq 0$, the left-hand side is at least $\sum_{j=1}^{n} c_j\,x_j$. Hence $\sum_{j=1}^{n} c_j\,x_j \leq \sum_{i=1}^{m} b_i\,\bar{\pi}_i$ for every feasible $\boldsymbol{x}$. □

- In the car example the optimal solution of D(LP) is $\tilde{\boldsymbol{\pi}}(\mathrm{D(LP)}) = (\frac{17}{62}, \frac{2}{31})$, and

$$z(\mathrm{D(LP)}) = 45 \cdot \frac{17}{62} + 49 \cdot \frac{2}{31} = \frac{765}{62} + \frac{98}{31} = \frac{31}{2} = z(\mathrm{LP}).$$

The other two vertices of the dual feasible region give the bounds $\frac{45}{2}$ and $\frac{147}{4}$.

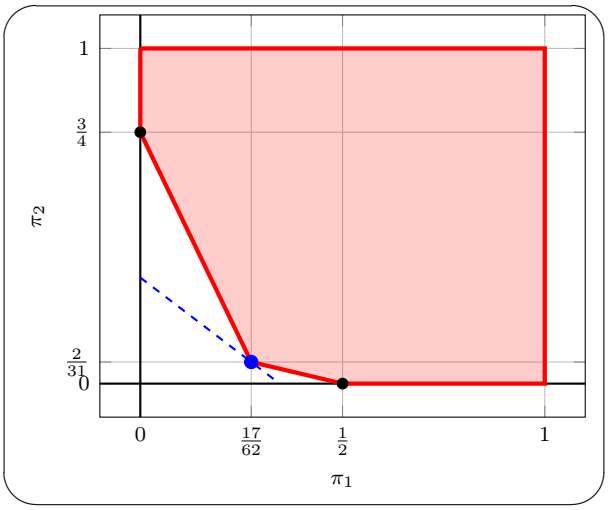


Figure 2.6: The feasible region of the dual LP problem (red, unbounded) and its optimal solution $(\frac{17}{62}, \frac{2}{31})$ (blue), with the objective line $45\,\pi_1 + 49\,\pi_2 = \frac{31}{2}$ (dashed). The other vertices (black) give the bounds $\frac{45}{2}$ and $\frac{147}{4}$.

Every feasible solution of the dual problem is a certificate of an upper bound, found without solving the LP relaxation. The best certificate is an optimal dual solution, and it has the same value as the LP relaxation.

2.5.1 Summary of the notation

- **ILP problem**: optimal solution $\tilde{\boldsymbol{x}}(\text{ILP})$, optimal value $z(\text{ILP})$, feasible solution $\bar{\boldsymbol{x}}$ with $\text{LB}(\bar{\boldsymbol{x}}) = \sum_j c_j \bar{x}_j \leq z(\text{ILP})$.
- **LP relaxation**: optimal solution $\tilde{\boldsymbol{x}}(\text{LP})$, optimal value $z(\text{LP})$, upper bound $z(\text{ILP}) \leq z(\text{LP})$.
- **Dual problem**: optimal solution $\tilde{\boldsymbol{\pi}}(\text{D(LP)})$, optimal value $z(\text{D(LP)}) = z(\text{LP})$, and $z(\text{ILP}) \leq \sum_i b_i \bar{\pi}_i$ for every dual feasible $\bar{\boldsymbol{\pi}}$.

2.6 Typical mistakes

Warning

- **Rounding the bound when it is not valid.** The bound $\lfloor z(\text{LP}) \rfloor$ requires all the variables and all the costs to be integer. In a MIP problem the bound is $z(\text{LP})$.
- **Rounding the solution and calling it optimal.** A rounded solution is feasible at best, and in general not optimal. It may even be infeasible, as the nearest integer point $(2, 5)$ of Example 2.2.
- **Using the value of a feasible LP solution as a bound.** Only the optimal value of the LP relaxation, or the value of a dual feasible solution, is an upper bound.
- **Multipliers with a negative sign.** The multipliers of constraints $\leq$ must be non-negative, and the combined coefficients must be at least the costs.
- **Deriving a bound on a variable from a constraint with negative coefficients.** The proof of (2.5) needs all the coefficients to be non-negative.
- **Believing that a stronger formulation always improves the bound.** The bounds $x_1 \leq 4$ and $x_2 \leq 3$ leave the bound 15 of the car example unchanged.

2.7 In summary

In summary

- The LP relaxation drops integrality, so $z(\text{ILP}) \leq z(\text{LP})$; with integer variables and integer costs the bound can be rounded down to $\lfloor z(\text{LP}) \rfloor$.
- Lower and upper bound define the optimality gap; optimality is certified when the gap is zero.
- Rounding an optimal LP solution gives a feasible solution at best, never a certificate.
- The upper bounds on the variables $x_j \leq \lfloor b_i / a_{ij} \rfloor$ come from constraints with non-negative coefficients, and they help only if they cut off the LP optimum.
- Non-negative multipliers of the constraints give upper bounds; the best ones solve the dual problem, whose optimal value is $z(\text{LP})$.

Dashboard of the car example. The bounds on $z(\text{ILP})$ found so far in this module:

Chapter	Lower bound	Upper bound	From
Integer linear programming	12, 14	–	heuristic solutions $(4, 0)$ and $(4, 1)$
LP relaxation and dual bounds	14	15	$\lfloor z(\text{LP}) \rfloor$, with $z(\text{LP}) = \frac{31}{2}$

Part II

Strengthening the formulation

Chapter 3

Ideal formulations

Prerequisites

- From this module, the chapters «Integer linear programming» and «LP relaxation and dual bounds»: the feasible region of an ILP problem, the LP relaxation and its bound, and the car example.
- From the module *Linear Programming*, Chapter 15 «Geometry of linear programming»: vertices of a polyhedron, and the fact that an LP problem with an optimal solution has an optimal vertex.
- From the module *Convex optimization*, Chapter 1 «Convex sets»: convex combinations and convex hulls. A short recap is given in this chapter.

Objectives

At the end of this chapter you will be able to:

- compute the convex hull of the feasible solutions of a small ILP problem, and classify its points as vertices, boundary points or interior points;
- define a formulation of an ILP problem, and decide whether it is ideal;
- explain why an LP problem over an ideal formulation solves the ILP problem;
- compare two formulations of the same problem through their LP relaxations;
- explain why ideal formulations exist but are hard to find.

Where we were

In the car example we have the lower bound 14 and the upper bound $\lfloor z(\text{LP}) \rfloor = 15$. The gap of $\frac{1}{14}$ is due to the polyhedron of the constraints, which contains fractional points that no integer point can justify. We ask: which polyhedron would give no gap at all?

3.1 Why ideal formulations?

- The same set of integer points can be described by many different systems of linear inequalities, and each of them gives a different LP relaxation with a different bound.
- The best polyhedron is the smallest one that contains all the integer points. If we could use it, the LP relaxation would return an integer solution, and the ILP problem would be as easy as an LP problem.
- This polyhedron is the **convex hull** of the feasible solutions, and a system of inequalities that describes it is an **ideal formulation**.

3.2 The convex hull

- Imagine a rubber band stretched around a set of points and then released: it settles around the smallest convex region containing the points. This is the convex hull.

Definition 3.1: of convex hull

Given a set $S \subseteq \mathbb{R}^n$, a **convex combination** of points $\boldsymbol{s}^1, \boldsymbol{s}^2, \ldots, \boldsymbol{s}^k \in S$ is a point $\sum_{h=1}^{k} \lambda_h \boldsymbol{s}^h$ with $\lambda_h \geq 0$ and $\sum_{h=1}^{k} \lambda_h = 1$. The **convex hull** of S, denoted by $\text{conv}(S)$, is the set of all the convex combinations of points of S.

- The convex hull is the smallest convex set containing S. If S is a finite set of points, the convex hull is a bounded polyhedron (a **polytope**), and its **vertices** are points of S. The other points of S lie either on the boundary of the polytope, or in its interior: they are not needed to describe it.
- For an ILP problem we take $S = F$, the set of the feasible solutions. The vertices of $\text{conv}(F)$ are integer feasible solutions.
- In the car example, the convex hull of the 16 integer points has 6 vertices:

$$(0,0), \quad (4,0), \quad (4,1), \quad (3,2), \quad (1,3), \quad (0,3).$$

Of the other 10 points, five lie on its boundary, namely $(0,1)$, $(0,2)$, $(1,0)$, $(2,0)$ and $(3,0)$, and five in its interior, namely $(1,1)$, $(2,1)$, $(3,1)$, $(1,2)$ and $(2,2)$.

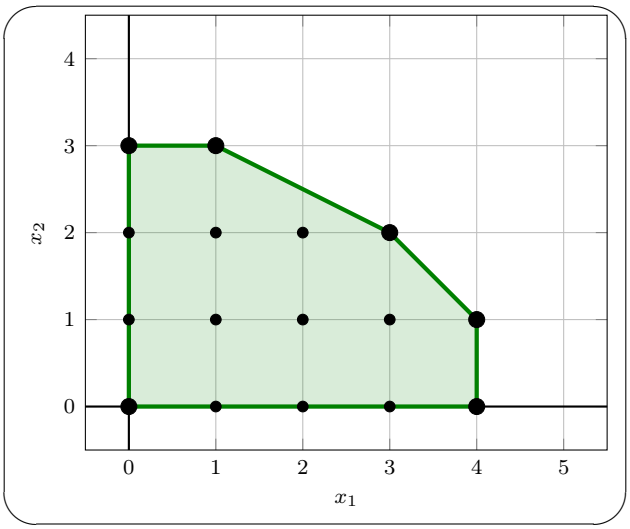


Figure 3.1: The convex hull of the 16 integer points of the car example (green). Its 6 vertices (black squares) are integer points; five other points lie on its boundary and five in its interior.

3.3 Formulations and ideal formulations

Definition 3.2: of formulation

A **formulation** of an ILP problem with feasible region F is a polyhedron $P = \{\boldsymbol{x} \in \mathbb{R}^n : \sum_j a_{ij}\, x_j \leq b_i,\ \forall i,\ x_j \geq 0,\ \forall j\}$ such that $P \cap \mathbb{Z}^n = F$. The **LP relaxation of the formulation** is the problem of maximizing $\sum_j c_j\, x_j$ over P.

- An ILP problem has many formulations: they share the integer points and differ in the fractional points. Since P is convex and contains F, every formulation contains the convex hull:

$$\text{conv}(F) \ \subseteq \ P.$$

Definition 3.3: of ideal formulation

A formulation P of an ILP problem is **ideal** if $P = \text{conv}(F)$, that is, if the feasible region of its LP relaxation is exactly the convex hull of the integer feasible solutions.

Theorem 3.4: LP over an ideal formulation

Let P be an ideal formulation of an ILP problem with $F \neq \emptyset$ and P bounded. Then the LP relaxation has an optimal vertex, every optimal vertex is an optimal solution of the ILP problem, and $z(\text{LP}) = z(\text{ILP})$.

Proof. The polyhedron $P = \text{conv}(F)$ is non-empty and bounded, so the LP problem has an optimal vertex of P. The vertices of $\text{conv}(F)$ belong to F, so this vertex is a feasible ILP solution, whose value is $z(\text{LP})$. Since $z(\text{ILP}) \leq z(\text{LP})$, the vertex is optimal for the ILP problem. □

- The simplex algorithm returns an optimal vertex, so with an ideal formulation it returns an integer solution. If the LP problem has several optimal solutions, a method that does not return vertices can give a fractional point on the optimal face: this is why the theorem speaks of an optimal vertex.

3.3.1 The car example

- The formulation of the car example is **not ideal**. The point $(\frac{7}{2}, \frac{5}{2})$ belongs to P but not to $\text{conv}(F)$, because it violates $x_1 + x_2 \leq 5$, which all the vertices of the hull satisfy: $\text{conv}(F) \subsetneq P$.

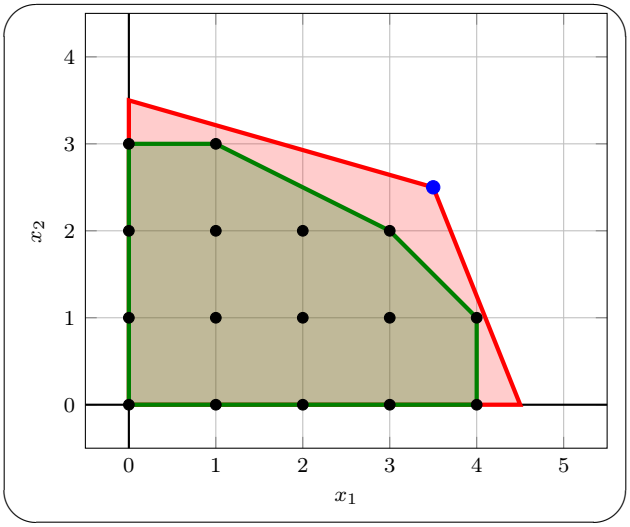


Figure 3.2: The car example: the polyhedron of the LP relaxation (red) contains the convex hull of the integer points (green), and the optimal solution of the LP relaxation (blue) is outside the hull.

- The convex hull is described by the following four inequalities and the non-negativity constraints:

$$
\begin{array}{rrcrcl}
\max & 3\,x_1 & + & 2\,x_2 & & \\
\text{subject to} & x_1 & + & x_2 & \leq & 5 \\
& x_1 & + & 2\,x_2 & \leq & 7 \\
& x_1 & & & \leq & 4 \\
& & & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$

- These inequalities define exactly the 6 vertices of the hull, and the same 16 integer points. The LP relaxation of this **ideal formulation** has the optimal solution $(4, 1)$, which is integer, with $z(\text{LP}) = 14 = z(\text{ILP})$.

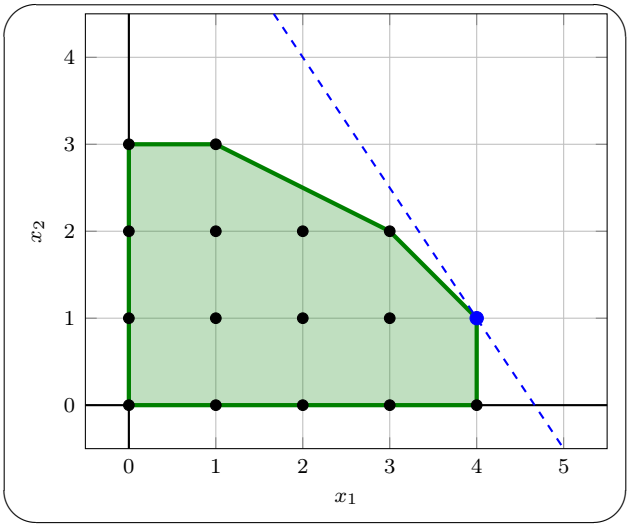


Figure 3.3: The ideal formulation of the car example: its feasible region (green) is the convex hull of the integer points. The optimal solution of the LP problem is the integer point $(4, 1)$ (blue).

3.4 Comparing formulations

- Two formulations P_1 and P_2 of the same ILP problem are compared through the polyhedra. If $P_1 \subseteq P_2$, we say that P_1 is **stronger** than P_2.
- A stronger formulation never gives a worse upper bound: the maximum over a smaller set is not larger, so

$$P_1 \subseteq P_2 \quad \Longrightarrow \quad z(\text{ILP}) \;\leq\; z(\text{LP}_1) \;\leq\; z(\text{LP}_2).$$

- The ideal formulation is the strongest of all, because it is contained in every formulation. In the car example the original formulation gives the bound $\lfloor \frac{31}{2} \rfloor = 15$, and the ideal one gives the bound 14.

Among the formulations of the same ILP problem, we prefer the ones with the smallest polyhedron: their LP relaxations give the best upper bounds, and the ideal formulation gives the optimal value.

3.5 Good news and bad news

- **Good news.** An ideal formulation exists for every ILP problem whose feasible region is bounded or, more in general, whose data are rational (Meyer's theorem; in this module the data are integer). The convex hull of the integer points is a polyhedron, so it is described by a finite number of linear inequalities.
- **Bad news.** Finding the ideal formulation is in general very difficult. It can require an exponential number of inequalities, and finding the necessary ones is often as hard as solving the ILP problem itself.
- The practical way out is to move step by step. We keep the formulation we have and add inequalities that are satisfied by all the integer points, and that cut off fractional points: they are the **valid inequalities** of the next chapter.

3.6 Typical mistakes

Warning

- **Believing that the points that are not vertices lie strictly inside the hull.** They can lie on its boundary, as $(0, 1)$ and $(3, 0)$ in the car example.
- **Checking only the optimal solution.** A formulation can return an integer optimal solution without being ideal: the LP optimum can be integer for one objective function and fractional for another one.
- **Believing that every formulation is ideal.** The formulation of the car example has the right integer points but contains fractional points outside the hull.
- **Adding inequalities that remove integer points.** A formulation must keep all the points of F: the polyhedron can shrink only down to the convex hull, never below it.

3.7 In summary

In summary

- The convex hull of the feasible solutions is the smallest polyhedron containing them, and its vertices are integer feasible solutions.
- A formulation is ideal if its polyhedron is the convex hull; then an optimal vertex of the LP relaxation is an optimal ILP solution.
- A stronger formulation, with a smaller polyhedron, never gives a worse bound.
- The car example has an ideal formulation with 4 inequalities, which gives $z(\text{LP}) = 14 = z(\text{ILP})$.
- Ideal formulations exist for rational data, but in general they need exponentially many inequalities and are hard to find.

Dashboard of the car example. The bounds on $z(\text{ILP})$ found so far in this module:

Chapter	Lower bound	Upper bound	From
Integer linear programming	12, 14	–	heuristic solutions $(4, 0)$ and $(4, 1)$
LP relaxation and dual bounds	14	15	$\lfloor z(\text{LP}) \rfloor$, with $z(\text{LP}) = \frac{31}{2}$
Ideal formulations	14	14	ideal formulation, optimal vertex $(4, 1)$

Chapter 4

Valid inequalities

Prerequisites

- From this module, the chapter «LP relaxation and dual bounds»: the LP relaxation, the rounded bound $\lfloor z(\text{LP}) \rfloor$, and bounds from non-negative multipliers of the constraints.
- From this module, the chapter «Ideal formulations»: formulations, the convex hull, and the comparison of formulations through their polyhedra.

Objectives

At the end of this chapter you will be able to:

- define a valid inequality, and say when it cuts off a fractional point;
- derive a valid inequality from non-negative multipliers of the constraints by rounding down;
- improve a valid inequality until it touches a feasible integer point;
- measure the effect of a valid inequality on the optimal value of the LP relaxation and on the upper bound;
- explain why valid inequalities are the practical way to approach an ideal formulation.

Where we were

In the car example the ideal formulation gives the upper bound 14, equal to the lower bound, so it certifies that $(4, 1)$ is optimal. But we found its four inequalities by looking at the convex hull, which is not possible in general. We now learn to **derive** such inequalities from the constraints.

4.1 Why valid inequalities?

- We cannot compute the ideal formulation of a large ILP problem, but we can add to the model a few inequalities that tighten the LP relaxation. Each one removes some fractional points and keeps all the integer ones.
- In the car example the upper bound is 15 and the lower bound is 14. Is there an inequality, satisfied by all the integer points, that pushes the upper bound down to 14?

4.2 Valid inequalities

Definition 4.1: of valid inequality

An inequality $\sum_{j=1}^{n} \alpha_j x_j \leq \beta$ is a **valid inequality** for an ILP problem with feasible region F if it is satisfied by all the points of F.

- A valid inequality is satisfied by all the integer feasible solutions, but it can be violated by some fractional points of the LP relaxation. We say that it **cuts off** these points.
- Adding a valid inequality to the formulation does not change F, and it shrinks the polyhedron P. The new formulation is stronger, so its LP relaxation gives an upper bound that is not worse.
- A valid inequality is satisfied by a point of F if and only if it is satisfied by the convex hull, so it is valid for $\text{conv}(F)$. The inequalities of the ideal formulation are valid inequalities.
- A valid inequality is **tight** if at least one point of F satisfies it with equality: its right-hand side cannot be decreased without losing a feasible solution.

A valid inequality removes fractional points and never an integer feasible solution. Adding valid inequalities makes the formulation stronger and the upper bound better.

- The inequalities $x_1 \leq 4$ and $x_2 \leq 3$ of the previous chapters are valid inequalities of the car example. They did not cut off the optimal solution of the LP relaxation, so they did not improve the bound.

4.3 Deriving valid inequalities by rounding

- We know how to combine the constraints with non-negative multipliers. In an ILP problem we can round the result, because the variables are integer.

Proposition 4.2: valid inequalities by rounding

Let $\pi_1, \pi_2, \dots, \pi_m \geq 0$ be multipliers such that $\sum_{i=1}^{m} \pi_i\, a_{ij} \geq \alpha_j$, with $\alpha_j \in \mathbb{Z}$, for all $j \in \{1, 2, \dots, n\}$. Then the inequality

$$\sum_{j=1}^{n} \alpha_j\, x_j \;\leq\; \left\lfloor \sum_{i=1}^{m} \pi_i\, b_i \right\rfloor \tag{4.1}$$

is valid for the ILP problem.

Proof. For every feasible solution, $\sum_j \alpha_j\, x_j \leq \sum_j \big(\sum_i \pi_i\, a_{ij}\big) x_j \leq \sum_i \pi_i\, b_i$, by $x_j \geq 0$ and by the constraints multiplied by $\pi_i \geq 0$. The left-hand side is an integer, because α_j and x_j are integer, so it is at most the integer part of the right-hand side. □

- The proposition needs all the variables to be integer and non-negative. With a continuous variable the left-hand side is not an integer, and the right-hand side cannot be rounded.

Example 4.1: rounding with multiplier $\frac{1}{2}$

In the car example, multiply the first constraint by $\frac{1}{2}$: we get $5\,x_1 + 2\,x_2 \leq \frac{45}{2}$. Since the left-hand side is integer, we obtain the valid inequality

$$5\,x_1 + 2\,x_2 \leq 22.$$

Multiplying the second constraint by $\frac{1}{2}$ we get $2\,x_1 + 7\,x_2 \leq \frac{49}{2}$, hence

$$2\,x_1 + 7\,x_2 \leq 24.$$

Both inequalities cut off the optimal solution $(\frac{7}{2}, \frac{5}{2})$ of the LP relaxation: $5 \cdot \frac{7}{2} + 2 \cdot \frac{5}{2} = \frac{45}{2} > 22$ and $2 \cdot \frac{7}{2} + 7 \cdot \frac{5}{2} = \frac{49}{2} > 24$. The new LP relaxations give:

Added inequality	$\tilde{\boldsymbol{x}}(\mathrm{LP})$	$z(\mathrm{LP})$	$\lfloor z(\mathrm{LP}) \rfloor$
none	$(\frac{7}{2}, \frac{5}{2})$	$\frac{31}{2}$	15
$5\,x_1 + 2\,x_2 \leq 22$	$(\frac{105}{31}, \frac{157}{62})$	$\frac{472}{31}$	15
$2\,x_1 + 7\,x_2 \leq 24$	$(\frac{219}{62}, \frac{75}{31})$	$\frac{957}{62}$	15

The LP optimum moves, but the upper bound does not improve: $\lfloor \frac{472}{31} \rfloor = \lfloor \frac{957}{62} \rfloor = 15$.

A valid inequality that cuts off the optimal LP solution does not always improve the upper bound. The choice of the multipliers matters: we look for inequalities that touch the integer points.

4.4 Improving a valid inequality

- Given the coefficients α_j, the strongest valid inequality has the smallest possible right-hand side, which is the maximum of $\sum_j \alpha_j x_j$ over F. We find it by lowering the right-hand side β of (4.1) until a feasible integer point reaches it.

Algorithm

Input: An ILP problem in canonical maximization form with $F \neq \emptyset$, and integer coefficients $\alpha_1, \alpha_2, \ldots, \alpha_n$.

Output: A tight valid inequality $\sum_{j=1}^{n} \alpha_j x_j \leq \beta$.

Step 0: **Coarse bound**

Find multipliers $\pi_i \geq 0$ with $\sum_i \pi_i a_{ij} \geq \alpha_j$ for all j, and set $\beta \leftarrow \lfloor \sum_i \pi_i b_i \rfloor$.

Step 1: **Test the right-hand side**

If no feasible integer point satisfies $\sum_j \alpha_j x_j = \beta$, set $\beta \leftarrow \beta - 1$ and repeat Step 1.

Step 2: **Tight inequality**

A feasible integer point satisfies $\sum_j \alpha_j x_j = \beta$, so $\sum_j \alpha_j x_j \leq \beta$ is valid and tight. `STOP`.

- In Step 0 the multipliers can be chosen freely: the better they are, the fewer repetitions of Step 1 are needed. In Step 1, for two variables, we substitute one variable using the equation and bound the other one with the constraints: if the bounds contradict each other, no point exists.

Example 4.2: the inequality $x_1 + x_2 \leq 5$

We look for a valid inequality with $\alpha = (1, 1)$.

- **Step 0.** With the multipliers $(\frac{1}{14}, \frac{1}{14})$ the sum of the constraints, divided by 14, is $x_1 + \frac{9}{7} x_2 \leq \frac{47}{7}$. The coefficients are at least $(1, 1)$, so $x_1 + x_2 \leq \lfloor \frac{47}{7} \rfloor = 6$, and we set $\beta = 6$.
- **Step 1, $\beta = 6$.** Let $x_1 + x_2 = 6$, so $x_2 = 6 - x_1$. The first constraint becomes $6\,x_1 + 24 \leq 45$, that is $x_1 \leq \frac{7}{2}$, hence $x_1 \leq 3$. The second becomes $84 - 10\,x_1 \leq 49$, that is $x_1 \geq \frac{7}{2}$, hence $x_1 \geq 4$. The bounds contradict each other: no feasible point has $x_1 + x_2 = 6$, and we set $\beta = 5$.
- **Step 1, $\beta = 5$.** Let $x_2 = 5 - x_1$. The constraints become $x_1 \leq \frac{25}{6}$, hence $x_1 \leq 4$, and $x_1 \geq \frac{21}{10}$, hence $x_1 \geq 3$. There is no contradiction, and the points $(3, 2)$ and $(4, 1)$ are feasible with $x_1 + x_2 = 5$.
- **Step 2.** The inequality $x_1 + x_2 \leq 5$ is valid and tight.

It cuts off $(\frac{7}{2}, \frac{5}{2})$ (the sum is 6). The new LP relaxation has the optimal solution $(\frac{25}{6}, \frac{5}{6})$ with $z(\text{LP}) = \frac{85}{6}$, so the upper bound is $\lfloor \frac{85}{6} \rfloor = 14$.

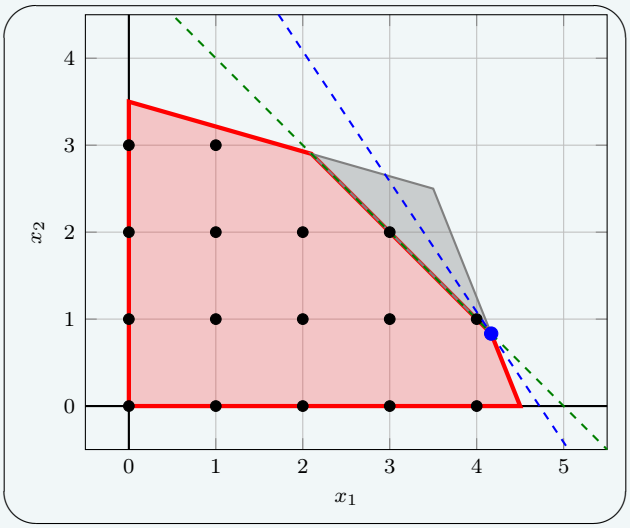


Figure 4.1: The valid inequality $x_1 + x_2 \leq 5$ (green) cuts off the gray triangle, which contains no integer point, and the optimal solution of the previous LP relaxation. The new optimal solution is $(\frac{25}{6}, \frac{5}{6})$ (blue), with $z = \frac{85}{6}$ (dashed line).

Example 4.3: the inequality $x_1 + 2\,x_2 \leq 7$

We look for a valid inequality with $\alpha = (1, 2)$.

- **Step 0.** With the multipliers $(\frac{3}{62}, \frac{4}{31})$ the combination of the constraints is $x_1 + 2\,x_2 \leq \frac{17}{2}$, because $\frac{3 \cdot 10}{62} + \frac{4 \cdot 4}{31} = 1$, $\frac{3 \cdot 4}{62} + \frac{4 \cdot 14}{31} = 2$ and $\frac{3 \cdot 45}{62} + \frac{4 \cdot 49}{31} = \frac{17}{2}$. Hence $x_1 + 2\,x_2 \leq \lfloor \frac{17}{2} \rfloor = 8$, and we set $\beta = 8$.
- **Step 1,** $\beta = 8$. Let $x_1 = 8 - 2\,x_2$. The first constraint becomes $80 - 16\,x_2 \leq 45$, hence $x_2 \geq 3$; the second becomes $32 + 6\,x_2 \leq 49$, hence $x_2 \leq 2$. They contradict each other, and we set $\beta = 7$.
- **Step 1,** $\beta = 7$. Let $x_1 = 7 - 2\,x_2$: the constraints become $70 - 16\,x_2 \leq 45$, hence $x_2 \geq 2$, and $28 + 6\,x_2 \leq 49$, hence $x_2 \leq 3$. The points $(3, 2)$ and $(1, 3)$ are feasible with $x_1 + 2\,x_2 = 7$.
- **Step 2.** The inequality $x_1 + 2\,x_2 \leq 7$ is valid and tight.

The new LP relaxation has the optimal solution $(\frac{31}{8}, \frac{25}{16})$ with $z(\text{LP}) = \frac{59}{4}$, so the upper bound is $\lfloor \frac{59}{4} \rfloor = 14$.

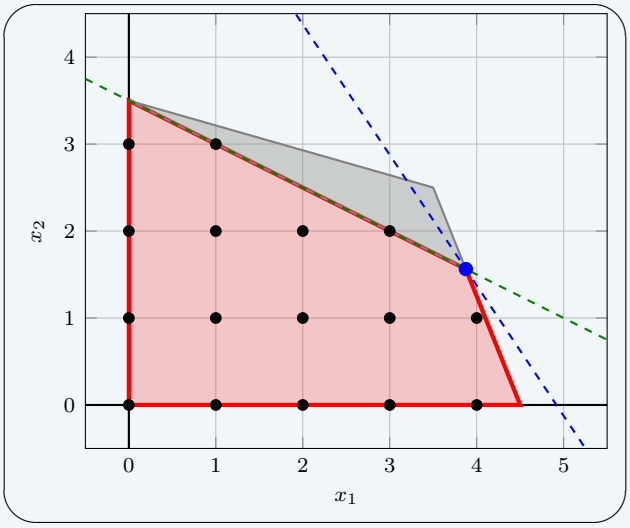


Figure 4.2: The valid inequality $x_1 + 2\,x_2 \leq 7$ (green) cuts off the gray triangle, which contains no integer point. The new optimal solution is $(\frac{31}{8}, \frac{25}{16})$ (blue), with $z = \frac{59}{4}$ (dashed line).

With the valid inequality $x_1 + x_2 \leq 5$, or with $x_1 + 2\,x_2 \leq 7$, the upper bound drops from 15 to 14. It equals the lower bound given by $(4, 1)$: **the solution** $(4, 1)$ **is certified optimal**, and the gap is zero.

- The two inequalities, together with $x_1 \leq 4$ and $x_2 \leq 3$, are exactly the four inequalities of the ideal formulation of the previous chapter. For the car example we have reached the convex hull with valid inequalities derived from the constraints.

4.5 What comes next

- By hand and with two variables we chose the directions α looking at the picture. For real problems we need two ingredients: valid inequalities for the structures that appear in the models, and a systematic way to find inequalities that cut off the current LP optimum.
- The first ingredient gives the families of inequalities for specific structures, such as the **cover inequalities** of the knapsack constraints (next chapter).
- The second ingredient is the rounding of this chapter, applied systematically to the rows of the optimal dictionary of the LP relaxation: the **Gomory fractional cuts** of the cutting plane algorithm, and then the **branch-and-cut** algorithm.

4.6 Typical mistakes

Warning

- **Rounding when a variable is continuous.** The rounding of (4.1) needs integer variables and integer coefficients α_j.
- **Rounding up instead of down.** The right-hand side is rounded down. Rounding up gives an inequality that is not valid, and it can remove integer solutions.
- **Believing that every valid inequality improves the bound.** The inequalities $5\,x_1 + 2\,x_2 \leq 22$ and $2\,x_1 + 7\,x_2 \leq 24$ cut off the LP optimum, but leave the bound at 15.
- **Using multipliers that are not large enough.** The combined coefficients must be at least α_j for every j, and the multipliers must be non-negative.
- **Stopping at a coarse right-hand side.** The inequality $x_1 + x_2 \leq 6$ is valid, but it does not cut off the LP optimum of the car example (its sum is 6); the tight inequality $x_1 + x_2 \leq 5$ does.

4.7 In summary

In summary

- A valid inequality is satisfied by all the feasible integer solutions; it can cut off fractional points of the LP relaxation, and it makes the formulation stronger.
- Non-negative multipliers of the constraints, with the right-hand side rounded down, give valid inequalities for ILP problems.
- Lowering the right-hand side until an integer point touches the line gives a tight inequality.
- A valid inequality helps only if it cuts off the LP optimum and improves the rounded bound.
- In the car example, $x_1 + x_2 \leq 5$ or $x_1 + 2\,x_2 \leq 7$ lowers the upper bound from 15 to 14, and certifies that $(4, 1)$ is optimal.

Dashboard of the car example. The bounds on $z(\text{ILP})$ found so far in this module:

Chapter	Lower bound	Upper bound	From
Integer linear programming	12, 14	–	heuristic solutions $(4, 0)$ and $(4, 1)$
LP relaxation and dual bounds	14	15	$\lfloor z(\text{LP}) \rfloor$, with $z(\text{LP}) = \frac{31}{2}$
Ideal formulations	14	14	ideal formulation, optimal vertex $(4, 1)$
Valid inequalities	14	14	$x_1 + x_2 \leq 5$ gives $z(\text{LP}) = \frac{85}{6}$

Chapter 5

Cover inequalities

Prerequisites

- From this module, the chapters «Integer linear programming» and «LP relaxation and dual bounds»: ILP problems, the LP relaxation, and the upper bound given by the LP relaxation.
- From this module, the chapters «Ideal formulations» and «Valid inequalities»: the convex hull of the integer solutions, valid inequalities, and cuts that remove fractional points.
- From the module *Combinatorial optimization*, the chapter «Knapsack problems»: the knapsack problem and the LP relaxation of the knapsack model, solved with Dantzig's rule.

Objectives

At the end of this chapter you will be able to:

- write the cover inequalities of a knapsack constraint, and prove that they are valid;
- list the covers and the minimal covers of an instance, and describe the cover closure;
- separate a cover inequality from a fractional point by solving a cover separation problem, exactly or with the greedy heuristic algorithm;
- strengthen a cover inequality into an extended cover inequality;
- compute the violation of a cover inequality and the distance of a point from it.

5.1 Knapsack constraints and their LP relaxation

- Given:
  - a knapsack of *capacity* $c \in \mathbb{Z}_{\geq 1}$,
  - a set $\mathscr{I} = \{1, 2, \ldots, n\}$ of n items,
  - for each item $j \in \mathscr{I}$, a *weight* $w_j \in \mathbb{Z}_{\geq 1}$,

  and a binary variable x_j for each item $j \in \mathscr{I}$, equal to 1 if item j is selected in the knapsack and to 0 otherwise, the *knapsack constraint* (also called *budget constraint*) is:

$$\sum_{j \in \mathscr{I}} w_j x_j \leq c, \tag{5.1a}$$

$$x_j \in \{0, 1\}, \qquad \forall j \in \mathscr{I}. \tag{5.1b}$$

- We assume that

$$w_j \leq c \quad \text{for all} \quad j \in \mathscr{I} \qquad \text{and} \qquad \sum_{j \in \mathscr{I}} w_j > c.$$

  The first assumption guarantees that every item can be selected, the second one that not all the items fit together into the knapsack.

- We denote by X the *knapsack set*, i.e., the set of feasible solutions of (5.1a)–(5.1b):

$$X = \left\{ \boldsymbol{x} \in \{0, 1\}^n : \sum_{j \in \mathscr{I}} w_j x_j \leq c \right\}. \tag{5.2}$$

  X is a finite set of binary points. Its convex hull $\text{conv}(X)$, called the *knapsack polytope*, is the smallest convex set containing X.

- An exact description of $\text{conv}(X)$ is generally hard to obtain. We therefore start from the LP relaxation

$$P = \left\{ \boldsymbol{x} \in [0, 1]^n : \sum_{j \in \mathscr{I}} w_j x_j \leq c \right\}. \tag{5.3}$$

  The aim is to strengthen P by removing fractional points without removing any point of X, thereby bringing the relaxation closer to $\text{conv}(X)$.

- One or several knapsack constraints appear in many ILP models. Valid inequalities for this set remain valid when further constraints are imposed.

- With profits p_j, maximizing $\sum_{j \in \mathscr{I}} p_j x_j$ over P gives an upper bound on the corresponding integer maximization problem; adding valid inequalities can improve this upper bound.

- **A challenge.** Before reading on, consider the instance of Example 5.1, with four items. The optimal solution of its LP relaxation is fractional. Can you find an inequality that cuts it off without removing any feasible binary point? The examples of this chapter answer this question.

5.2 Covers and cover inequalities

Definition 5.1: Cover and collection of all covers

A *cover* is a subset $C \subseteq \mathscr{I}$ whose total weight exceeds the knapsack capacity. The collection of all covers is

$$\mathcal{C} = \left\{ C \subseteq \mathscr{I} : \sum_{j \in C} w_j > c \right\}. \tag{5.4}$$

Since weights and capacity are integer, the condition is equivalent to $\sum_{j \in C} w_j \geq c + 1$.

Definition 5.2: Cover inequality

Given $C \in \mathcal{C}$, the associated *cover inequality* is

$$\sum_{j \in C} x_j \leq |C| - 1. \tag{5.5}$$

Proposition 5.3

For every cover $C \in \mathcal{C}$, the cover inequality (5.5) is valid for X.

Proof. Suppose that a point $\boldsymbol{x} \in X$ violates (5.5). The sum of $|C|$ binary variables is then greater than $|C| - 1$, hence $x_j = 1$ for every $j \in C$. Since all weights are positive,

$$\sum_{j \in \mathscr{I}} w_j x_j \geq \sum_{j \in C} w_j > c.$$

This contradicts the capacity constraint: at most $|C| - 1$ items of C can be selected together. □

5.2.1 Minimal covers

Definition 5.4: Minimal cover

A cover C is *minimal* if no proper subset of C is a cover. Equivalently,

$$\sum_{j \in C} w_j > c, \qquad \sum_{j \in C} w_j - w_i \leq c \quad \forall i \in C. \tag{5.6}$$

- Since weights are positive, it suffices to remove one item at a time: if each of these $|C|$ subsets fits into the knapsack, every smaller subset fits as well.

Proposition 5.5

The inequalities of the minimal covers, together with the upper bounds $x_j \leq 1$ for all $j \in \mathscr{I}$, imply every cover inequality.

Proof. Starting from a cover C, remove items as long as the remaining set is still a cover: this gives a minimal cover $D \subseteq C$. Adding the cover inequality of D and the upper bounds of the items in $C \setminus D$ gives

$$\sum_{j \in C} x_j = \sum_{j \in D} x_j + \sum_{j \in C \setminus D} x_j \leq |D| - 1 + |C \setminus D| = |C| - 1. \qquad \square$$

5.2.2 The cover closure

Definition 5.6: Cover closure

The *cover closure* of P is the polyhedron

$$P^{\text{cov}} = \left\{ \boldsymbol{x} \in P : \sum_{j \in C} x_j \leq |C| - 1 \quad \forall C \in \mathcal{C} \right\}. \tag{5.7}$$

- Since all cover inequalities are valid,

  $$\text{conv}(X) \subseteq P^{\text{cov}} \subseteq P,$$

  and in general the first inclusion is strict.
- By Proposition 5.5, the inequalities of the minimal covers are sufficient to describe P^{cov}.
- The number of covers can be exponential in n. For example, if $w_j = 1$ for all $j \in \mathscr{I}$ and $c = 1$, every subset with at least two items is a cover, i.e., there are $2^n - n - 1$ covers. Also the minimal covers can be exponentially many: for $w_j = 1$ for all $j \in \mathscr{I}$, $n = 2m$ and $c = m$, they are the $\binom{2m}{m+1}$ subsets of $m + 1$ items.
- Therefore, cover inequalities are not added all at once: they are added only when they are violated, i.e., they are *separated* (Section 5.4).

Example 5.1: A Knapsack Problem

Consider $\mathscr{I} = \{1, 2, 3, 4\}$, capacity $c = 9$, and the following profits and weights.

j	1	2	3	4
p_j	6	5	8	9
w_j	2	3	6	7

The associated LP_{KP} model is:

$$\begin{array}{rrcrcrcrcl} \max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\ \text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

The profit-to-weight ratios p_j/w_j are $3, 5/3, 4/3, 9/7$, in non-increasing order. Dantzig's rule inserts the items in this order, and inserts fractionally the first item that does not fit: it gives an optimal solution of $\mathrm{LP_{KP}}$. An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP_{KP}})$ is:

$$\tilde{x}_1 = \tilde{x}_2 = 1, \qquad \tilde{x}_3 = \frac{2}{3}, \qquad \tilde{x}_4 = 0.$$

The optimal value $z(\mathrm{LP_{KP}})$ is $6 + 5 + 8(2/3) = 49/3$.

Example 5.2: The cover closure of the four-item instance

The instance of Example 5.1 has the following seven covers.

C	$\sum_{j\in C} w_j$	$\lvert C\rvert$	Cover inequality	Minimal
$\{2,4\}$	10	2	$x_2 + x_4 \leq 1$	yes
$\{3,4\}$	13	2	$x_3 + x_4 \leq 1$	yes
$\{1,2,3\}$	11	3	$x_1 + x_2 + x_3 \leq 2$	yes
$\{1,2,4\}$	12	3	$x_1 + x_2 + x_4 \leq 2$	no
$\{1,3,4\}$	15	3	$x_1 + x_3 + x_4 \leq 2$	no
$\{2,3,4\}$	16	3	$x_2 + x_3 + x_4 \leq 2$	no
$\{1,2,3,4\}$	18	4	$x_1 + x_2 + x_3 + x_4 \leq 3$	no

Only the first three covers are minimal; their inequalities already describe the cover closure. Denote by $\mathrm{LP^{cov}}$ the LP obtained by adding them to $\mathrm{LP_{KP}}$. An optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP^{cov}})$ is:

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = \frac{3}{5}, \qquad \tilde{x}_3 = \frac{2}{5}, \qquad \tilde{x}_4 = \frac{2}{5}.$$

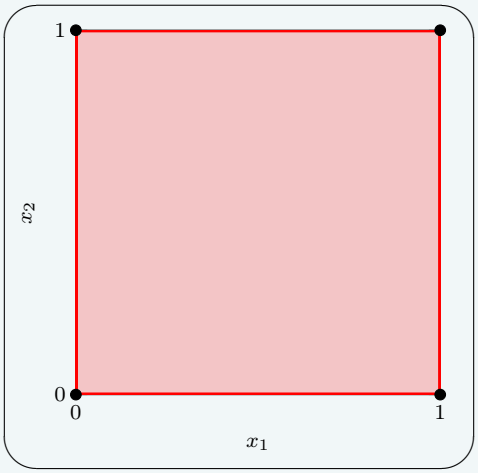
The optimal value is $z(\mathrm{LP^{cov}}) = 79/5 < z(\mathrm{LP_{KP}}) = 49/3$.

5.3 Geometric interpretation

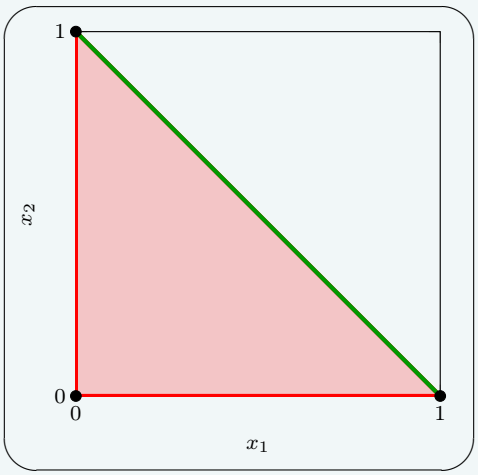
- All coordinates in the figures are the actual variable values. The ticks mark the coordinates of the blue point and of the main vertices; a few vertices have no tick, to keep the axes readable. In 3D, dashed edges are hidden behind the visible faces.
- Every step is illustrated both in two and in three dimensions, with the same colours in all figures: the red region is the continuous region under consideration; green identifies a cover inequality (an edge in 2D, a face in 3D); black points are the feasible binary points; the blue point is the fractional point discussed in the text.
- First consider only the domain, $[0,1]^2$ or $[0,1]^3$: no capacity constraint is imposed in the next two examples.

Example 5.3: A cover inequality in the unit square

In $[0,1]^2$, the inequality $x_1 + x_2 \leq 1$ removes only the binary point $(1,1)$ and preserves the binary points $(0,0)$, $(1,0)$ and $(0,1)$.



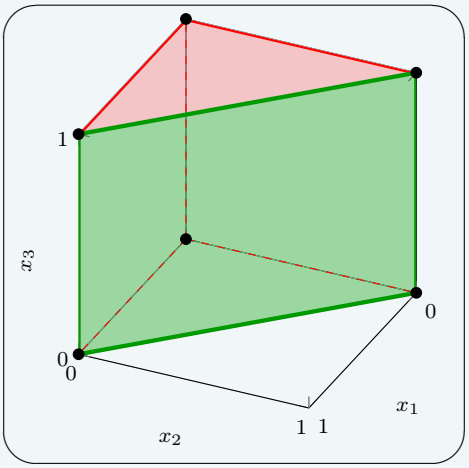
Unit square: domain only.



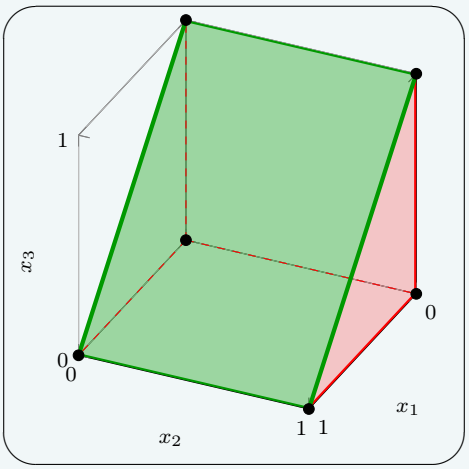
Domain with $x_1 + x_2 \leq 1$.

Example 5.4: Cover inequalities in the unit cube

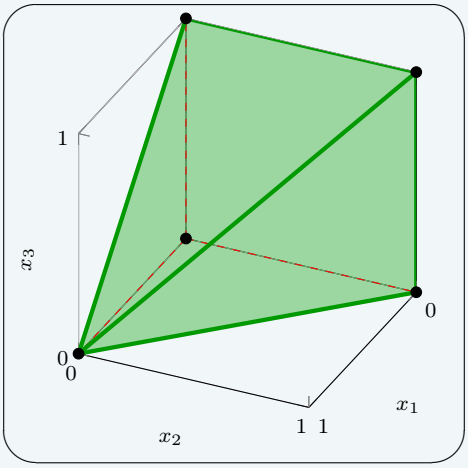
- In $[0,1]^3$, the inequality $x_1 + x_2 \leq 1$ is independent of x_3; its boundary is parallel to the x_3 axis. Similarly, the boundary of $x_1 + x_3 \leq 1$ is parallel to the x_2 axis.
- The two halfspaces remove different portions of the cube. Their intersection preserves exactly the binary points $(0,0,0), (1,0,0), (0,1,0), (0,0,1), (0,1,1)$.
- The inequality $x_1 + x_2 + x_3 \leq 2$ removes only the binary point $x_1 = x_2 = x_3 = 1$.



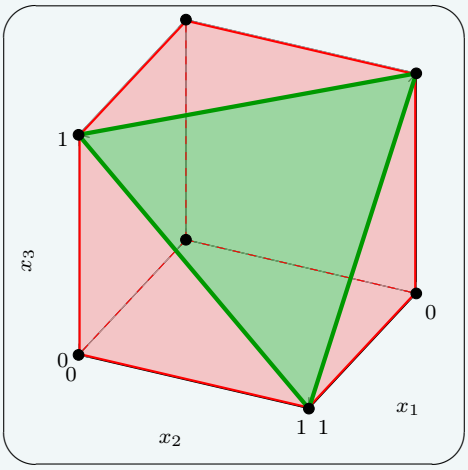
Domain with $x_1 + x_2 \leq 1$.



Domain with $x_1 + x_3 \leq 1$.



Domain with both inequalities.



Domain with $x_1 + x_2 + x_3 \leq 2$.

Example 5.5: A cover in two dimensions

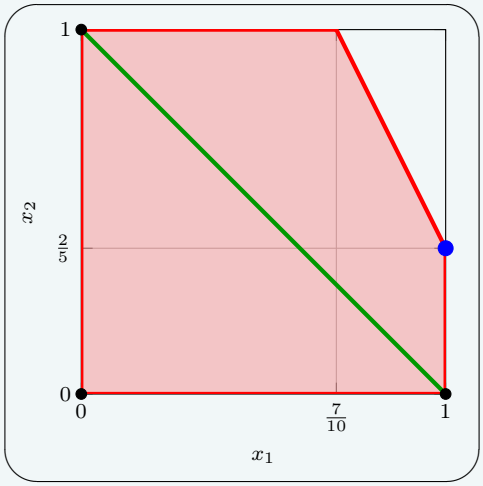
For $\mathscr{I} = \{1, 2\}$, consider

$$P_2 = \{\boldsymbol{x} \in [0,1]^2 : 10x_1 + 5x_2 \leq 12\}.$$

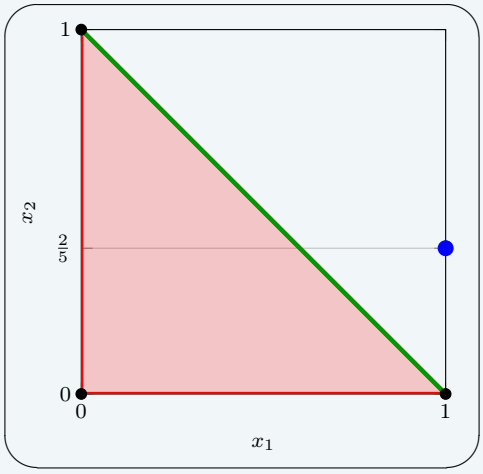
The cover $C = \{1, 2\}$ gives $x_1 + x_2 \leq 1$. The point $\tilde{x}_1 = 1$, $\tilde{x}_2 = 2/5$ satisfies the capacity constraint at equality, but violates the cover by $2/5$. The original polygon has vertices

$$(0,0),\ (1,0),\ (1,2/5),\ (7/10,1),\ (0,1).$$

After adding the cut, the region is the triangle with vertices $(0,0), (1,0), (0,1)$. Thus all three feasible binary points are preserved, and the capacity constraint becomes redundant, since $10x_1 + 5x_2 \leq 10 < 12$ on the triangle.



Original P_2 and the boundary $x_1 + x_2 = 1$.



$P_2 \cap \{x_1 + x_2 \leq 1\}$: the blue point is removed.

Example 5.6: Capacity and successive cover cuts

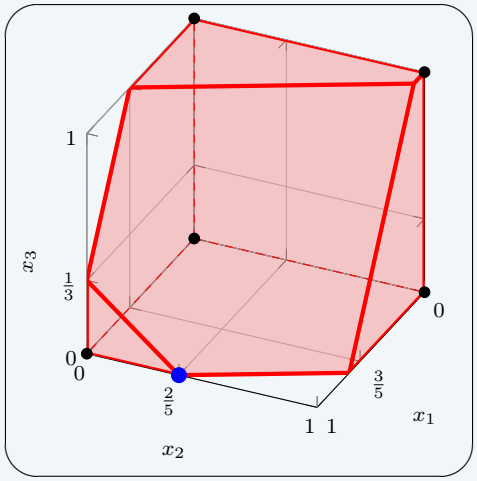
For $\mathscr{I} = \{1, 2, 3\}$, let

$$P_A = \{\boldsymbol{x} \in [0,1]^3 : 10x_1 + 5x_2 + 6x_3 \leq 12\}.$$

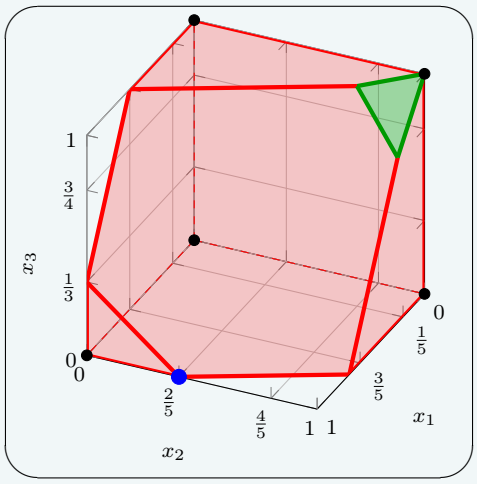
There are exactly three covers:

$$\begin{aligned} \{1,2,3\} &: x_1 + x_2 + x_3 \leq 2, \\ \{1,3\} &: x_1 + x_3 \leq 1, \\ \{1,2\} &: x_1 + x_2 \leq 1. \end{aligned}$$

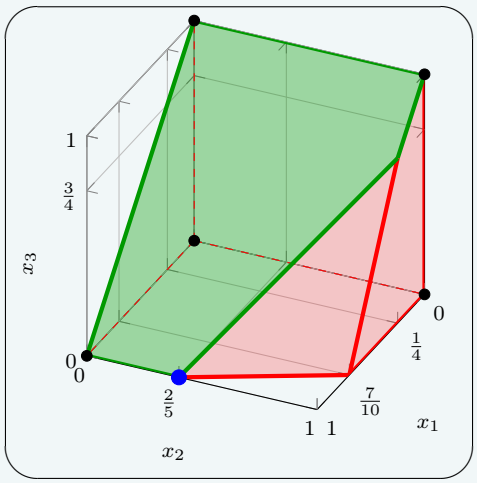
Only the last two are minimal. The blue point $\tilde{x}_1 = 1$, $\tilde{x}_2 = 2/5$, $\tilde{x}_3 = 0$ lies in P_A. It satisfies the first inequality and is on the boundary of the second; the third is violated by $2/5$, exactly as in Example 5.5. Unlike the previous domain-only panels, every following panel includes the capacity constraint.



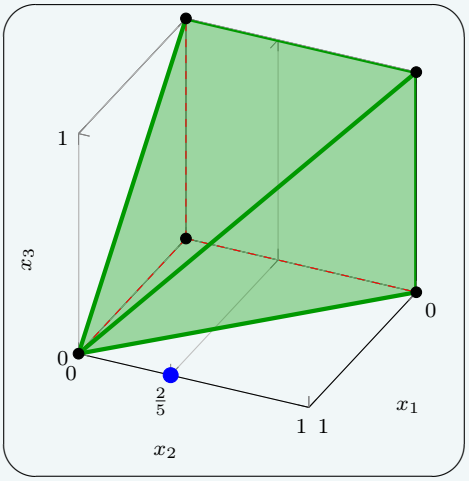
Original P_A.



Add $x_1 + x_2 + x_3 \leq 2$.



Also add $x_1 + x_3 \leq 1$.



Also add $x_1 + x_2 \leq 1$: the cover closure.

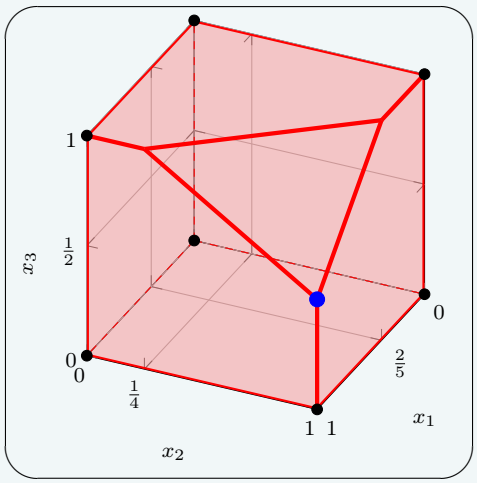
- For the objective $\max 40x_1 + 15x_2 + 12x_3$, the optimal solution over P_A is $\tilde{\boldsymbol{x}} = (1, 2/5, 0)$, with value 46. Over the cover closure, the optimal solution is $(1, 0, 0)$, with value 40, which is also optimal for the KP.
- In this instance the cover closure coincides with $\text{conv}(X)$: its vertices, shown in the last panel, are all binary. This is a property of this instance, not of every cover closure.

Example 5.7: A different capacity constraint

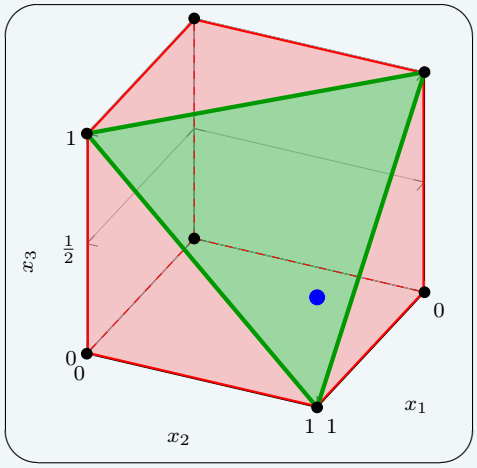
For $\mathscr{I} = \{1, 2, 3\}$, consider now the instance

$$P_B = \{\boldsymbol{x} \in [0, 1]^3 : 5x_1 + 4x_2 + 6x_3 \leq 12\}.$$

Every pair fits, whereas the three items together have weight 15. Thus the only cover is $\{1, 2, 3\}$ and it is minimal. Its inequality is $x_1 + x_2 + x_3 \leq 2$. The blue point $\tilde{x}_1 = \tilde{x}_2 = 1$, $\tilde{x}_3 = 1/2$ has weight 12 and violates this cut by $1/2$.



Original P_B.



$P_B \cap \{x_1 + x_2 + x_3 \leq 2\}$.

- The capacity plane intersects the cube edges at $(1, 1, 1/2)$, $(1, 1/4, 1)$ and $(2/5, 1, 1)$.
- The cover removes these fractional vertices while preserving all seven feasible binary points. As in the 2D example, after the cut the capacity constraint is redundant: the sum of the two largest weights is $6 + 5 = 11 < 12$.
- The polyhedron $P_B \cap \{x_1 + x_2 + x_3 \leq 2\}$ has only binary vertices (last panel): here a single cover inequality gives $\text{conv}(X)$.

5.4 Separation of cover inequalities

Given a point $\tilde{\boldsymbol{x}} \in P$, the **cover separation problem** (CSP) consists in finding a cover inequality violated by $\tilde{\boldsymbol{x}}$, or in proving that $\tilde{\boldsymbol{x}}$ satisfies all the cover inequalities.

Formally, the CSP asks to find a cover

$$C \in \mathcal{C}: \qquad \sum_{j \in C} \tilde{x}_j > |C| - 1,$$

or to prove that

$$\sum_{j \in C} \tilde{x}_j \leq |C| - 1 \qquad \forall C \in \mathcal{C}.$$

- Typically, $\tilde{\boldsymbol{x}}$ is an optimal solution of the current LP relaxation, but any point of P can be separated.
- For a cover C, the cover inequality is violated by $\tilde{\boldsymbol{x}}$ if

$$\sum_{j \in C} \tilde{x}_j > |C| - 1. \tag{5.8}$$

Moving $\sum_{j \in C} \tilde{x}_j$ to the right-hand side and 1 to the left-hand side, and writing $|C| =$

$\sum_{j \in C} 1$, we obtain the equivalent conditions:

$$
\begin{aligned}
|C| - \sum_{j \in C} \tilde{x}_j &< 1, \\
\sum_{j \in C} 1 - \sum_{j \in C} \tilde{x}_j &< 1, \\
\sum_{j \in C} (1 - \tilde{x}_j) &< 1.
\end{aligned} \tag{5.9}
$$

- Assigning to each item j the cost $q_j = 1 - \tilde{x}_j \geq 0$, the CSP amounts to finding a cover of *minimum cost*: a violated cover inequality exists if and only if this minimum cost is less than 1.

5.4.1 An ILP formulation for the CSP

- Introduce n binary variables z_j: $z_j = 1$ if item j belongs to the cover and $z_j = 0$ otherwise.
- The objective minimizes the total cover cost; the single weight constraint forces the selected set to be a cover.

$$
\begin{aligned}
(\mathrm{ILP}_{\mathrm{CSP}}) \qquad \min \quad & \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j) z_j && && \text{(5.10a)} \\
\text{subject to} \quad & \sum_{j \in \mathscr{I}} w_j z_j \geq c + 1, && && \text{(5.10b)} \\
& z_j \in \{0, 1\}, && \forall j \in \mathscr{I}. && \text{(5.10c)}
\end{aligned}
$$

Proposition 5.7

A cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\mathrm{ILP}_{\mathrm{CSP}}) < 1$.

Proof. Every feasible assignment of the binary variables z_j represents a cover $C = \{j : z_j = 1\}$, and its objective is $\sum_{j \in C} (1 - \tilde{x}_j)$. By (5.9), this is less than 1 exactly when the corresponding cut is violated. Taking the minimum over all covers proves both implications. The CSP is feasible because $\mathscr{I}$ itself is a cover. □

Let $\tilde{z}_1, \tilde{z}_2, \ldots, \tilde{z}_n$ be an optimal solution of $\mathrm{ILP}_{\mathrm{CSP}}$. If $z(\mathrm{ILP}_{\mathrm{CSP}}) < 1$, the cover

$$
\tilde{C} = \{ j \in \mathscr{I} : \ \tilde{z}_j = 1 \}
$$

is associated with the cover inequality $\sum_{j \in \tilde{C}} x_j \leq |\tilde{C}| - 1$ maximally violated by $\tilde{\boldsymbol{x}}$. Otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$.

- To find a violated cover inequality, it is not necessary to solve $\mathrm{ILP}_{\mathrm{CSP}}$ to optimality: any feasible solution $\bar{z}_1, \ldots, \bar{z}_n$ of cost less than 1 gives the cover $\bar{C} = \{j \in \mathscr{I} : \ \bar{z}_j = 1\}$, whose cover inequality is violated by $\tilde{\boldsymbol{x}}$.
- Conversely, a feasible solution of cost at least 1 does *not* prove that no violated cover inequality exists: this requires the optimal value (or a lower bound) of $\mathrm{ILP}_{\mathrm{CSP}}$.

- If $z(\text{ILP}_{\text{CSP}}) < 1$, the maximum violation of a cover inequality at $\tilde{\boldsymbol{x}}$ is $1 - z(\text{ILP}_{\text{CSP}})$.

Example 5.8: Separation of the cover inequalities

We consider again Example 5.1. The cover inequalities are separated iteratively: at each iteration, the current LP is solved, the ILP_{CSP} is solved for its optimal solution $\tilde{\boldsymbol{x}}$, and the maximally violated cover inequality is added, until no cover inequality is violated.

Iteration 1. An optimal solution of LP_{KP} is

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = 1, \qquad \tilde{x}_3 = \frac{2}{3}, \qquad \tilde{x}_4 = 0, \qquad z(\text{LP}_{\text{KP}}) = \frac{49}{3}.$$

The associated ILP_{CSP} is:

$$\begin{array}{rrcrcrcrcl} \min & & & & & \frac{1}{3}\,z_3 & + & z_4 & & \\ \text{subject to} & 2\,z_1 & + & 3\,z_2 & + & 6\,z_3 & + & 7\,z_4 & \geq & 10 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

An optimal solution of ILP_{CSP} is $\tilde{z}_1 = \tilde{z}_2 = \tilde{z}_3 = 1$, $\tilde{z}_4 = 0$, with value $z(\text{ILP}_{\text{CSP}}) = \frac{1}{3} < 1$. Accordingly, $\tilde{C} = \{j \in \mathscr{J} : \tilde{z}_j = 1\} = \{1, 2, 3\}$ is the cover associated with the cover inequality maximally violated by $\tilde{\boldsymbol{x}}$:

$$x_1 + x_2 + x_3 \leq 2, \qquad \text{since} \qquad \tilde{x}_1 + \tilde{x}_2 + \tilde{x}_3 = 1 + 1 + \frac{2}{3} = \frac{8}{3} > 2.$$

Iteration 2. Adding this cover inequality to LP_{KP}, we obtain the following LP, denoted LP_1:

$$\begin{array}{rrcrcrcrcl} \max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\ \text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\ & x_1 & + & x_2 & + & x_3 & & & \leq & 2 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

An optimal solution of LP_1 is

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = 1, \qquad \tilde{x}_3 = 0, \qquad \tilde{x}_4 = \frac{4}{7}, \qquad z(\text{LP}_1) = \frac{113}{7}.$$

The associated ILP_{CSP} is:

$$\begin{array}{rrcrcrcrcl} \min & & & & & z_3 & + & \frac{3}{7}\,z_4 & & \\ \text{subject to} & 2\,z_1 & + & 3\,z_2 & + & 6\,z_3 & + & 7\,z_4 & \geq & 10 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

An optimal solution of ILP_{CSP} is $\tilde{z}_1 = 0$, $\tilde{z}_2 = 1$, $\tilde{z}_3 = 0$, $\tilde{z}_4 = 1$, with value $z(\text{ILP}_{\text{CSP}}) = \frac{3}{7} < 1$ (the cover $\{1, 2, 4\}$ has the same cost; we choose the minimal cover). Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{z}_j = 1\} = \{2, 4\}$ and the maximally violated cover inequality is

$$x_2 + x_4 \leq 1, \qquad \text{since} \qquad \tilde{x}_2 + \tilde{x}_4 = 1 + \frac{4}{7} = \frac{11}{7} > 1.$$

Iteration 3. Adding this cover inequality to LP_1, we obtain the following LP, denoted LP_2:

$$\begin{array}{rrcrcrcrcl}
\max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & + & x_3 & & & \leq & 2 \\
& & & x_2 & & & + & x_4 & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution of LP_2 is

$$\tilde{x}_1 = 1, \qquad \tilde{x}_2 = \frac{3}{5}, \qquad \tilde{x}_3 = \frac{2}{5}, \qquad \tilde{x}_4 = \frac{2}{5}, \qquad z(\text{LP}_2) = \frac{79}{5}.$$

The associated ILP_{CSP} is:

$$\begin{array}{rrcrcrcrcl}
\min & & & \frac{2}{5}\,z_2 & + & \frac{3}{5}\,z_3 & + & \frac{3}{5}\,z_4 & & \\
\text{subject to} & 2\,z_1 & + & 3\,z_2 & + & 6\,z_3 & + & 7\,z_4 & \geq & 10 \\
& z_1, & & z_2, & & z_3, & & z_4 & \in & \{0, 1\}
\end{array}$$

An optimal solution of ILP_{CSP} is $\tilde{z}_1 = 0$, $\tilde{z}_2 = 1$, $\tilde{z}_3 = 0$, $\tilde{z}_4 = 1$, with value $z(\text{ILP}_{\text{CSP}}) = 1$ (the covers $\{1, 2, 3\}$ and $\{1, 2, 4\}$ have the same cost): **no** cover inequality is violated by $\tilde{\boldsymbol{x}}$, and the separation stops.

The final value $z(\text{LP}_2) = \frac{79}{5}$ coincides with the value $z(\text{LP}^{\text{cov}})$ of the cover closure (Example 5.2): the separation reaches the bound of the cover closure by adding only 2 of the 7 cover inequalities.

The solution $\tilde{\boldsymbol{x}}$ is still **fractional**, although it satisfies all the cover inequalities: the cover closure does not coincide, in general, with the convex hull of the integer solutions.

5.5 Transforming the CSP into a Knapsack Problem

- Consider a feasible assignment of the CSP variables z_j. Define the complementary binary variables y_j by
$$y_j = 1 - z_j \qquad \forall j \in \mathscr{I}.$$
Thus $y_j = 1$ if item j is outside the cover, whereas $y_j = 0$ if item j belongs to the cover.

- The weight constraint becomes
$$\sum_{j\in\mathscr{I}} w_j(1-y_j) \geq c+1 \quad \Longleftrightarrow \quad \sum_{j\in\mathscr{I}} w_j y_j \leq \sum_{j\in\mathscr{I}} w_j - (c+1).$$
By the assumption $\sum_{j\in\mathscr{I}} w_j > c$, this capacity is non-negative.

- The objective can be written as
$$\sum_{j\in\mathscr{I}}(1-\tilde{x}_j)(1-y_j) = n - \sum_{j\in\mathscr{I}} \tilde{x}_j - \sum_{j\in\mathscr{I}}(1-\tilde{x}_j)y_j.$$
Since the first two terms are constant, minimizing it is equivalent to the following KP, called the *complementary KP*, whose profits $1-\tilde{x}_j$ are non-negative (possibly zero).

$$\begin{aligned}
(\mathrm{ILP}^2_{\mathrm{CSP}}) \qquad \max \quad & \sum_{j\in\mathscr{I}}(1-\tilde{x}_j)y_j && && \text{(5.11a)}\\
\text{subject to} \quad & \sum_{j\in\mathscr{I}} w_j y_j \leq \sum_{j\in\mathscr{I}} w_j - (c+1), && && \text{(5.11b)}\\
& y_j \in \{0,1\}, && \forall j \in \mathscr{I}. && \text{(5.11c)}
\end{aligned}$$

- Complementation is a bijection between feasible solutions, so
$$z(\mathrm{ILP}_{\mathrm{CSP}}) = n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}). \tag{5.12}$$

Let $\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_n$ be an optimal solution of $\mathrm{ILP}^2_{\mathrm{CSP}}$. If $n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) < 1$, the cover
$$\tilde{C} = \{j \in \mathscr{I} : \ \tilde{y}_j = 0\}$$
is associated with the cover inequality $\sum_{j\in\tilde{C}} x_j \leq |\tilde{C}| - 1$ maximally violated by $\tilde{\boldsymbol{x}}$. Otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$.

- Any feasible solution $\bar{y}_1, \ldots, \bar{y}_n$ of $\mathrm{ILP}^2_{\mathrm{CSP}}$ gives the cover $\bar{C} = \{j \in \mathscr{I} : \ \bar{y}_j = 0\}$, with cost $n - \sum_{j\in\mathscr{I}} \tilde{x}_j - \sum_{j\in\mathscr{I}}(1-\tilde{x}_j)\bar{y}_j$: if this cost is less than 1, the cover inequality of $\bar{C}$ is violated by $\tilde{\boldsymbol{x}}$.

5.5.1 A greedy separation heuristic

The following greedy procedure constructs feasible values $\bar{y}_j$ for the complementary KP.

1. Set $\bar{y}_j = 0$ for all $j \in \mathscr{I}$ and the residual capacity $r = \sum_{j\in\mathscr{I}} w_j - (c+1)$. Sort the items by non-increasing efficiency $q_j/w_j = (1-\tilde{x}_j)/w_j$, breaking ties by increasing index.
2. Scan the items in this order. If item j has positive profit $q_j > 0$ and $w_j \le r$, set $\bar{y}_j = 1$ and $r := r - w_j$; otherwise skip it.
3. Reconstruct the cover $C = \{j : \bar{y}_j = 0\}$ and evaluate its cost. If desired, reduce it to a minimal cover by removing items as long as its weight exceeds c.

- Items with zero profit ($\tilde{x}_j = 1$) are never selected: they remain in the cover, which may then be non-minimal.
- Removing an item from a cover decreases its cost by $q_j \ge 0$: reducing a violated cover to a minimal one cannot decrease its violation.
- The greedy algorithm is a *heuristic* for the complementary KP: if it does not find a violated cover inequality, we cannot conclude that none exists.

Example 5.9: Complementary KP and greedy scan

For the point $\tilde{\boldsymbol{x}} = (1, 1, \frac{2}{3}, 0)$ of Iteration 1 in Example 5.8,

$$4 - \sum_{j\in\mathscr{I}} \tilde{x}_j = 4 - 1 - 1 - \frac{2}{3} = \frac{4}{3},$$

$$\sum_{j\in\mathscr{I}} w_j - (c+1) = 2 + 3 + 6 + 7 - (9+1) = 8.$$

The complementary KP is

$$\begin{array}{rccccccccc} \max & & & & & \frac{1}{3}\,y_3 & + & y_4 & & \\ \text{subject to} & 2\,y_1 & + & 3\,y_2 & + & 6\,y_3 & + & 7\,y_4 & \le & 8 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

j	q_j	w_j	q_j/w_j	Decision	Residual
4	1	7	$\frac{1}{7}$	take	1
3	$\frac{1}{3}$	6	$\frac{1}{18}$	skip	1
1	0	2	0	zero profit	1
2	0	3	0	zero profit	1

The scan gives $\bar{y}_1 = \bar{y}_2 = \bar{y}_3 = 0$ and $\bar{y}_4 = 1$, with profit 1. Items 3 and 4 cannot fit together; the only positive-profit alternatives have values 1/3 and 1. Thus this greedy solution is also optimal, and

$$z(\mathrm{ILP}_{\mathrm{CSP}}) = 4 - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{4}{3} - 1 = \frac{1}{3}.$$

The cover is $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1, 2, 3\}$, as found in Iteration 1 of Example 5.8.

5.6 Extended cover inequalities

- A cover inequality can be strengthened by also considering the items outside the cover that are at least as heavy as every item of the cover. Let

$$\widehat{w}(C) = \max_{j \in C} w_j$$

be the maximum weight of the items in C.

Definition 5.8: Extended cover inequality

For a cover C, let

$$E(C) = C \cup \{j \in \mathscr{I} \setminus C : w_j \geq \widehat{w}(C)\}.$$

The associated *extended cover inequality* is

$$\sum_{j \in E(C)} x_j \leq |C| - 1. \tag{5.13}$$

The right-hand side depends on $|C|$, not on $|E(C)|$.

Proposition 5.9

For every cover $C \in \mathcal{C}$, the extended cover inequality (5.13) is valid for X.

Proof. Let $k = |C|$ and take any set $S \subseteq E(C)$ with $|S| = k$. The sets $S \setminus C$ and $C \setminus S$ have the same number of items, and each item of $S \setminus C$ weighs at least $\widehat{w}(C)$, i.e., at least as much as each item of $C \setminus S$. Replacing the items of $C \setminus S$ with those of $S \setminus C$ gives

$$\sum_{j \in S} w_j \geq \sum_{j \in C} w_j > c.$$

Hence no point $\boldsymbol{x} \in X$ can select k items of $E(C)$: at most $k-1$ of them can be selected together. □

- Since $C \subseteq E(C)$ and the right-hand side is the same, the extended cover inequality dominates the cover inequality of C.
- It can be seen as a *lifting* of the coefficients of the items in $E(C) \setminus C$ from 0 to 1. It does not forbid a heavy item to be selected together with items of C: it limits to $|C| - 1$ the number of selected items of the whole set $E(C)$. General lifting techniques are outside the scope of these notes.

Example 5.10: Strengthening the four-item cover

In Example 5.1, consider the cover $C = \{1, 2, 3\}$. Then $\widehat{w}(C) = 6$, and item 4 has weight $7 \geq 6$, so

$$E(C) = \{1, 2, 3, 4\}, \qquad x_1 + x_2 + x_3 + x_4 \leq 2.$$

The strengthened LP, denoted LP_{ext}, is

$$
\begin{array}{rrcrcrcrcl}
\max & 6\,x_1 & + & 5\,x_2 & + & 8\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 6\,x_3 & + & 7\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & + & x_3 & + & x_4 & \leq & 2 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

An optimal solution of $\mathrm{LP}_{\mathrm{ext}}$ is $\tilde{\boldsymbol{x}} = (1, 0, 0, 1)$, with value $z(\mathrm{LP}_{\mathrm{ext}}) = 6 + 9 = 15$. Since it is binary, $\tilde{\boldsymbol{x}}$ is also optimal for the KP. We have $z(\mathrm{LP}_{\mathrm{ext}}) = 15 < z(\mathrm{LP}^{\mathrm{cov}}) = 79/5$: in this instance, one extended cover inequality is stronger than all the cover inequalities together.

5.7 Violation and distance from a cover inequality

- Let $C \subseteq \mathscr{I}$ be a cover. Its cover inequality is

$$\sum_{j \in C} x_j \leq |C| - 1,$$

  and its boundary hyperplane is

$$H_C = \left\{ \boldsymbol{x} \in \mathbb{R}^n : \sum_{j \in C} x_j = |C| - 1 \right\}.$$

- The violation of the cover inequality at a point $\tilde{\boldsymbol{x}}$ is

$$v_C(\tilde{\boldsymbol{x}}) = \max \left\{ 0, \sum_{j \in C} \tilde{x}_j - (|C| - 1) \right\}. \tag{5.14}$$

  The point violates the cover inequality precisely when $v_C(\tilde{\boldsymbol{x}}) > 0$.

- The coefficients of the cover inequality are 1 for $j \in C$ and 0 otherwise, so their Euclidean norm is $\sqrt{|C|}$. Therefore, the Euclidean distance of $\tilde{\boldsymbol{x}}$ from H_C is

$$d(\tilde{\boldsymbol{x}}, H_C) = \frac{\left| \sum_{j \in C} \tilde{x}_j - (|C| - 1) \right|}{\sqrt{|C|}}. \tag{5.15}$$

  If the cover inequality is violated,

$$d(\tilde{\boldsymbol{x}}, H_C) = \frac{v_C(\tilde{\boldsymbol{x}})}{\sqrt{|C|}}.$$

- The orthogonal projection $\boldsymbol{x}^{\perp}$ of $\tilde{\boldsymbol{x}}$ onto H_C has components

$$x_j^{\perp} = \begin{cases} \tilde{x}_j - \dfrac{\sum_{k \in C} \tilde{x}_k - (|C| - 1)}{|C|}, & j \in C, \\ \tilde{x}_j, & j \in \mathscr{I} \setminus C. \end{cases} \tag{5.16}$$

  Indeed, substitution gives $\sum_{j \in C} x_j^{\perp} = |C| - 1$.

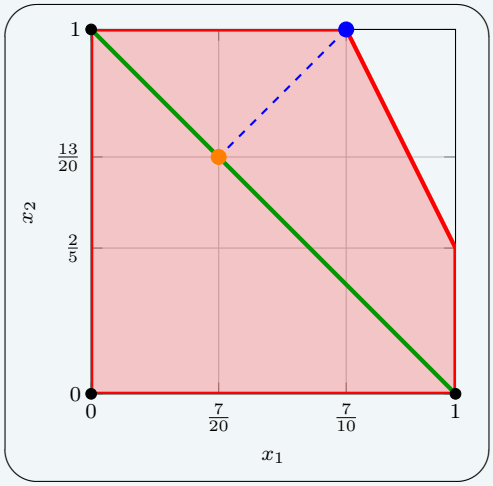
- The CSP maximizes the violation, not the distance: when the covers have different cardinalities, the two criteria can rank them differently. For example, at the point $\tilde{\boldsymbol{x}} = (1/10, 1, 1)$ of P_A (Example 5.6), the covers $\{1,2\}$, $\{1,3\}$ and $\{1,2,3\}$ all have violation $1/10$, but their distances are $1/(10\sqrt{2})$, $1/(10\sqrt{2})$ and $1/(10\sqrt{3})$, respectively.

Example 5.11: Orthogonal projection in two dimensions

Take the vertex $\tilde{x}_1 = 7/10$, $\tilde{x}_2 = 1$ of P_2 in Example 5.5 and the cover $C = \{1,2\}$. Its cover inequality is $x_1 + x_2 \geq 1$ and $|C| = 2$. Then

$$v_C(\tilde{\boldsymbol{x}}) = \frac{7}{10}, \qquad x_1^{\perp} = \frac{7}{20}, \qquad x_2^{\perp} = \frac{13}{20}, \qquad d(\tilde{\boldsymbol{x}}, H_C) = \frac{7}{10\sqrt{2}}.$$

The red region is P_2 and the green edge is the boundary H_C. The blue dashed segment is perpendicular to H_C and the orange point is the orthogonal projection.

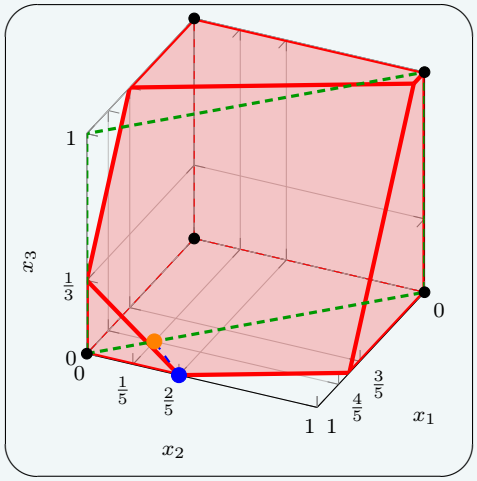


Example 5.12: Orthogonal projection in three dimensions

Take the point $\tilde{x}_1 = 1$, $\tilde{x}_2 = 2/5$, $\tilde{x}_3 = 0$ of P_A in Example 5.6 and the cover $C = \{1,2\}$. Its cover inequality is $x_1 + x_2 \geq 1$ and $|C| = 2$. Then

$$v_C(\tilde{\boldsymbol{x}}) = \frac{2}{5}, \qquad x_1^{\perp} = \frac{4}{5}, \qquad x_2^{\perp} = \frac{1}{5}, \qquad x_3^{\perp} = 0, \qquad d(\tilde{\boldsymbol{x}}, H_C) = \frac{2}{5\sqrt{2}}.$$

The red region is P_A and the green dashed rectangle is H_C restricted to the unit cube. Since $3 \notin C$, the projection does not change $\tilde{x}_3$.



5.8 Typical mistakes

Warning

- **Using a set that is not a cover.** The inequality $\sum_{j \in C} x_j \leq |C| - 1$ is valid only if the total weight of C exceeds the capacity c; check the weights before writing it.
- **Using $|E(C)| - 1$ for an extended cover.** The right-hand side of the extended cover inequality is $|C| - 1$, where C is the cover, not $|E(C)| - 1$.
- **Removing all the zero-profit items from a cover.** To obtain a minimal cover, remove items one at a time as long as the remaining set is still a cover: removing all of them may leave a set that is not a cover.
- **Stopping the separation at a heuristic failure.** If the greedy heuristic algorithm finds no violated cover inequality, we cannot conclude that none exists: only the optimal value of the separation problem proves it.
- **Confusing violation and distance.** The cover separation problem maximizes the violation; covers with the same violation can have different distances from the point.
- **Believing that the cover closure is the convex hull.** A point can satisfy all the cover inequalities and still be fractional (Example 5.8).

5.9 In summary

In summary

- A cover is a set of items whose total weight exceeds the capacity; its cover inequality allows at most $|C| - 1$ of its items, and it is valid for the knapsack set.
- The minimal covers are enough: their inequalities imply all the others, but they can be exponentially many, so they are separated only when violated.
- Separating a cover inequality means finding a cover of minimum cost $\sum_{j \in C}(1 - \tilde{x}_j)$: a violated inequality exists if and only if the minimum cost is less than 1; this is a knapsack problem.
- An extended cover inequality dominates the cover inequality of the same cover.
- The cover closure is closer to $\text{conv}(X)$ than the LP relaxation, but in general it is not equal to it.

Dashboard of the four-item instance. The lower and upper bounds on the optimal value of the knapsack problem of Example 5.1. The profits are integer, so the upper bound is the rounded-down value of the LP relaxation.

Source	Lower bound	Upper bound	From
LP relaxation	11	16	greedy solution $(1,1,0,0)$, $\lfloor z(\text{LP}_{\text{KP}}) \rfloor$ with $z(\text{LP}_{\text{KP}}) = \frac{49}{3}$
Cover closure	11	15	$\lfloor z(\text{LP}^{\text{cov}}) \rfloor$ with $z(\text{LP}^{\text{cov}}) = \frac{79}{5}$, 2 separated covers
Extended cover	15	15	$z(\text{LP}_{\text{ext}}) = 15$, optimal solution $(1,0,0,1)$